

Context-Dependent Modification of PFKFB3 in Hematopoietic Stem Cells Promotes Anaerobic Glycolysis and Ensures Stress Hematopoiesis

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Abstract

Metabolic pathways are plastic and rapidly change in response to stress or perturbation. Current metabolic profiling techniques require lysis of many cells, complicating the tracking of metabolic changes over time after stress in rare cells such as hematopoietic stem cells (HSCs). Here, we aimed to identify the key metabolic enzymes that define differences in glycolytic metabolism between steady-state and stress conditions in HSCs and elucidate their regulatory mechanisms. Through quantitative ¹³C metabolic flux analysis of glucose metabolism using high-sensitivity glucose tracing and mathematical modeling, we found that HSCs activate the glycolytic rate-limiting enzyme phosphofructokinase (PFK) during

proliferation and oxidative phosphorylation (OXPHOS) inhibition. Real-time measurement of adenosine triphosphate (ATP) levels in single HSCs demonstrated that proliferative stress or OXPHOS inhibition led to accelerated glycolysis via increased activity of PFKFB3, the enzyme regulating an allosteric PFK activator, within seconds to meet ATP requirements. Furthermore, varying stresses differentially activated PFKFB3 via PRMT1-dependent methylation during proliferative stress and via AMPK-dependent phosphorylation during OXPHOS inhibition. Overexpression of *Pfkfb3* induced HSC proliferation and promoted differentiated cell production, whereas inhibition or loss of *Pfkfb3* suppressed them. This study reveals the flexible and multilayered regulation of HSC glycolytic metabolism to sustain hematopoiesis under stress and provides techniques to better understand the physiological metabolism of rare hematopoietic cells.

Key Points

- Combined isotope tracing, mathematical modeling, and single cell ATP analysis enable high-resolution evaluation of blood cell metabolism.
- Under stress, HSCs quickly accelerate glycolysis to meet ATP demands and maintain hematopoiesis via context-dependent PFKFB3 activation.

eLife assessment

This **important** study provides novel strategies to overcome certain limitations when investigating the metabolism of hematopoietic stem cells, mainly due to their low abundance. The study provides **compelling** evidence suggesting that proliferative hematopoietic stem cells mainly use glycolysis (rather than mitochondrial OXPHOS or TCA cycle) as their primary energy source during emergency hematopoiesis. The article provides direct links between metabolic features and cell proliferation and explores alternative energy sources, and is of great interest to stem cell biologists.

Introduction

Activities governing nutrient requirements and metabolic pathways in individual cells maintain tissue homeostasis and respond to stress through metabolite production. Adenosine triphosphate (ATP), produced via cytosolic glycolysis and mitochondrial oxidative phosphorylation (OXPHOS), is the universal energy currency of all organisms; it regulates all anabolic or catabolic cellular activities^{1–3}. Precise control of intracellular ATP concentrations is critical, as ATP is the rate determiner of many ATP-dependent biochemical reactions^{4–9}.

Hematopoietic stem cells (HSCs) are tissue stem cells at the apex of the hematopoietic hierarchy; their function is maintained throughout life by a rigorous metabolic program and a complex interplay of gene expression, epigenetic regulation, intracellular signaling, chromatin remodeling, autophagy, and environmental factors^{10–14}. Conventional analyses of the metabolic programs of hematopoietic stem and progenitor cells (HSPCs) have revealed diverse differentiation potentials and cell-cycling statuses and coordinated activities that maintain hematopoiesis^{15–26}. Among the HSPC fractions, HSCs possess unique cell cycle quiescence, high self-renewal and differentiation capacity in response to stimuli, and resistance to cellular stress, including reactive oxygen species and physiological aging^{10,27–30}. These properties are regulated by a balance

between glycolysis and mitochondrial OXPHOS, requiring biosynthesis of ATP and various macromolecules that confer resilience to stress³¹. Among the known regulators of ATP-producing pathways, glycolytic enzymes maintain HSCs and hematopoietic progenitor cells (HPCs) by regulating cellular survival and cell cycle quiescence^{32–34}. Loss of mitochondrial genes in HPSCs also induces HSC differentiation defects^{35–37}. Moreover, disrupting the mitochondrial complex III subunit depletes both differentiated hematopoietic cells and quiescent HSCs²¹. Although glycolysis and the tricarboxylic acid (TCA) cycle are metabolically linked, pyruvate dehydrogenase kinase activity, which can uncouple these pathways, is required to maintain HSC function^{32,38}.

During HSC division, cell metabolism is reprogrammed to activate fatty acid β -oxidation (FAO) and purine metabolism^{39–41}. Furthermore, Liang et al. reported that activated HSCs mainly rely on glycolysis as their energy source⁴². However, the mechanisms by which each ATP-producing pathway and their connections are differentially regulated between HSCs and differentiated cells at steady state, during cell cycling, or during stress remain unknown. Recently, it has been shown that deeply quiescent HSCs do not activate cell cycle under stress^{43–45}. Therefore, it remains unclear whether metabolic changes such as the individual ATP-producing pathways and their interconnections occur uniformly in all HSCs, including these deeply quiescent HSCs. Furthermore, the underlying hub metabolic enzyme responsible for changes in the metabolic system of HSCs under stress has not been identified. HSCs are essential for cell therapy, including HSC transplantation, and in order to comprehensively elucidate the metabolic systems that have attracted attention as their regulatory mechanisms, recent studies have included metabolomic analyses using rare cell types such as HSCs^{23,46–49}, as well as isotope tracer analyses of undifferentiated hematopoietic cells purified after *in vivo* administration of isotopic glucose⁵⁰. Although these approaches are useful for obtaining comprehensive information on intracellular metabolites, they are not suited to track real-time changes in cellular metabolism at high resolution. Therefore, new approaches are necessary to analyze metabolites quantitatively and continuously without disturbing the physiological states of single cells while integrating the recently reported metabolome analysis techniques. In this study, we aimed to identify the key metabolic enzymes that define differences in glycolytic metabolism between steady-state and stress conditions in HSCs and elucidate their regulatory mechanisms using a quantitative and mathematical approach. Our findings provide a platform for quantitative metabolic analysis of rare cells such as HSCs, characterize the overall metabolic reprogramming of HSCs during stress loading, and highlight the key enzyme involved in this process.

Data sharing

All relevant data are available from the corresponding author upon reasonable request.

Results

HSC cell cycling increases anaerobic glycolytic flux

To determine how cell cycle progression alters HSC metabolism *in vivo*, we intraperitoneally and intravenously treated mice with 5-fluorouracil (5-FU) to induce HSC cell cycling (Supplemental Figure 1A). For analysis after 5-FU administration, the Lineage (Lin)[−] Sca-1⁺ c-Kit⁺ (LSK) gate was expanded to include HSCs with decreased c-Kit expression levels early after 5-FU treatment, for example high Sca-1-expressing cells and c-Kit-high to -dim Lin[−] cells, based on the previous report^{51,41} (Supplemental Figure 1B). This expanded LSK gate was consistent with the patterns of c-Kit expression observed in endothelial protein C receptor (EPCR)⁺ Lin[−] CD150⁺ CD48[−] cells (Supplemental Figure 1C) with high stem cell activity after 5-FU administration⁴¹. We observed a transient decrease in the number of quiescent HSCs (Ki67[−]) and an increase in the number of cell-cycling HSCs (Ki67⁺) on day 6 after 5-FU treatment (Supplemental Figure 1D). Along with the loss of cell quiescence, ATP concentration in HSCs decreased transiently on day 6 (Supplemental Figure

1E). Because the route of administration of 5-FU (intraperitoneal or intravenous) made no difference in the Ki67 positivity rate of HSCs (Supplemental Figure 1F), we administered 5-FU intraperitoneally for remaining experiments. Two methods were used to test whether cell cycle progression of HSCs after 5-FU treatment depends on the expression of EPCR. First, phosphorylation of Rb (pRb), a marker of cell cycle progression⁵², was analyzed in HSCs after 5-FU treatment. Analysis of EPCR⁺ and EPCR⁻ HSCs showed increased pRb in HSCs from 5-FU-treated mice in both fractions compared to HSCs from phosphate-buffered saline (PBS)-treated mice, regardless of EPCR expression (Supplemental Figure 1G-H). Second, we used a G₀ marker mouse line⁵³. These mice expressed a fusion protein of the p27 inactivation mutant p27K^{*} and the fluorescent protein mVenus (G₀ marker), allowing prospective identification of G₀ cells. We tested whether the expression of G₀ marker in HSCs was altered after 5-FU administration to the G₀ marker mice (Supplemental Figure 1I) and found that 5-FU treatment reduced the frequency of G₀ marker-positive HSCs, regardless of the EPCR expression (Supplemental Figure 1J-K). This was not observed in the PBS group. These results indicated that 5-FU administration induced cell cycle progression of entire HSCs in mice.

HSC cell cycling is preceded by the activation of intracellular ATP-related pathways that metabolize extracellular nutrients, including glucose^{39,40}, which are utilized in both ATP-producing and -consuming pathways, determining cellular ATP levels. Therefore, we examined the metabolic flux of glucose by performing *in vitro* IC-MS tracer analysis with uniformly carbon-labeled (U-¹³C₆) glucose to determine the pathways driving changes in ATP in 5-FU-treated HSCs (Figure 1A; Table S2). To avoid metabolite changes, samples were continuously chilled on ice during cell preparation, and the process from euthanasia to cell preparation was performed in the shortest possible time (see “Preparation and storage of *in vitro* U-¹³C₆-glucose tracer samples” section under “Methods” for more information). We found that changes in metabolite levels before and after sorting were present but limited (Supplemental Figure 2A). This result is consistent with the finding that the cell purification process does not significantly affect metabolite levels when sufficient care is taken in cell preparation⁵⁰. In 5-FU-treated HSCs, the levels of glycolytic metabolites derived from U-¹³C₆-glucose were double those observed in PBS-treated HSCs (Figure 1B-C; Supplemental Figure 2B). The total levels of TCA cycle intermediates derived from U-¹³C₆-glucose were similar between PBS- and 5-FU-treated cells (Figure 1D; Supplemental Figure 2B). Levels of U-¹³C₆-glucose-derived intermediates involved in the pentose phosphate pathway (PPP) and nucleic acid synthesis (NAS) were two-fold higher in 5-FU-treated than in PBS-treated HSCs, whereas no significant differences in the levels of metabolites were observed between both groups (Figure 1E-F; Supplemental Figure 2B). Notably, the labeling rate of metabolites during the first half of glycolysis was almost 100% in both groups, allowing us to easily track the labeled metabolites (Supplemental Figure 2C-E). This was thought to be due to the rapid replacement of unlabeled metabolites with labeled metabolites during exposure to U-¹³C₆-glucose because of the generally rapid glycolytic reaction. Conversely, the labeling rate of TCA cycle intermediates was consistently lower than that of glycolysis and PPP (Supplemental Figure 2D), suggesting that PBS- and 5-FU-treated HSCs prefer anaerobic glycolysis over aerobic glycolysis. To directly compare the metabolic systems of PBS- or 5-FU-treated HSCs, we conducted a Mito stress test using a Seahorse flux analyzer. Compared to PBS-treated HSCs, 5-FU-treated HSCs exhibited a higher extracellular acidification rate (ECAR), while their oxygen consumption rate (OCR) remained equal to that of PBS-treated HSCs (Figure 1G-H; Supplemental Figure 3A-B). After oligomycin treatment, PBS- and 5-FU-treated HSCs showed an increase in ECAR, suggesting a flexible activation of glycolysis upon OXPHOS inhibition (Figure 1G; Supplemental Figure 3A). Meanwhile, a decrease in OCR was more clearly observed in the 5-FU-treated HSCs (Figure 1H; Supplemental Figure 3B). Next, we evaluated whether glucose uptake in HSCs after 5-FU administration was differentially affected by the expression of EPCR. The fluorescent analog of glucose, 2-(N-(7-nitrobenz-2-oxa-1,3-diazol-4-yl)amino)-2-deoxyglucose (2-NBDG), was administered intravenously to mice⁵⁰ and its uptake in EPCR⁺ and EPCR⁻ HSCs was assayed (Figure 1I). Regardless of the EPCR expression, the 2-NBDG uptake was greater in HSCs treated with 5-FU than in those treated with PBS (Figure 1J-L). Increased 2-NBDG uptake in 5-FU-treated

HSCs was also observed in an *in vitro* 2-NBDG assay (Supplemental Figure 1L). Notably, even in the PBS-treated group, HSCs with high NBDG uptake were more proliferative than those with low NBDG uptake, similar to the state of HSCs after 5-FU administration (Supplemental Figure 1M). After 5-FU administration, there was an overall shift of the population from the G₀ to G₁ phase and a correlation between NBDG uptake and cell cycle progression was also observed (Supplemental Figure 1M). In both PBS- and 5-FU-treated groups, the marked variation in glucose utilization depending on the cell cycle suggests a direct link between HSC proliferation and increased glycolytic activity. Furthermore, compared to HSCs cultured under the quiescence-maintaining conditions of HSC achieved by hypoxia, abundant fatty acids, and low cytokines as we previously reported⁵⁴, HSCs cultured under cytokine-rich proliferative conditions were more resistant to the inhibition of OXPHOS by oligomycin (Supplementary Fig. 1N; Table S1). Overall, the results showed that 5-FU-treated HSCs exhibited activated glycolytic flux, increasing the turnover of ATP. Moreover, glycolytic flux into mitochondria was equally unchanged in PBS- and 5-FU-treated-HSCs, supporting that 5-FU activated anaerobic glycolysis in HSCs.

OXPHOS-inhibited HSCs exhibit compensatory glycolytic flux

Previous studies using mouse models of mitochondrial disease or defects in genes involved in electron transport chain and OXPHOS suggest that mitochondrial energy production is essential for maintaining HSC function^{21,35,37}, as is the glycolytic system. However, there have been no quantitative reports on how OXPHOS-inhibited HSCs can adapt their metabolism. To understand HSC metabolism under OXPHOS inhibition, we performed *in vitro* U-¹³C₆-glucose tracer analysis of oligomycin-treated HSCs (Figure 2A; Table S3). Similar to 5-FU-treated HSCs (Figure 1), oligomycin-treated HSCs exhibited glycolytic system activation (Figure 2B-C; Supplemental Figure 2B). Metabolite flux to the TCA cycle and PPP was unchanged, but flux to the NAS was significantly reduced in oligomycin-treated HSCs compared to that in steady-state HSCs (Figure 2D-F; Supplemental Figure 2B). The results suggested that OXPHOS-inhibited HSCs activated compensatory glycolytic flux and suppressed NAS flux. As with 5-FU-treated HSCs, analysis of oligomycin-treated HSCs also showed almost 100% labeling of metabolites in the first half of glycolysis (Supplemental Figure 2F-H), allowing us to easily track the labeled metabolites. To further validate the compensatory glycolytic activation of HSCs under OXPHOS inhibition, a Mito Stress test was performed on HSCs and other differentiated myeloid progenitors (MyPs, Lin⁻ Sca-1⁻ c-Kit⁺ (LKS⁻) cells). The results showed that ECAR were elevated in HSCs after oligomycin treatment compared to before oligomycin treatment (Figure 2G; Supplemental Figure 3C). No increase in ECAR was observed in MyPs (Figure 2G; Supplemental Figure 3C), supporting that inhibition of OXPHOS activated anaerobic glycolysis specifically in HSCs. Meanwhile, in HSCs, the decrease in OCR after oligomycin administration was less evident compared to MyPs (Figure 2H; Supplemental Figure 3D). In MyPs, both ECAR and OCR were downregulated (Figure 2G-H; Supplemental Figure 3C-D).

Phosphofructokinase (PFK) metabolism in HSCs is activated during proliferation and OXPHOS inhibition

To investigate whether glycolytic activation in HSCs after 5-FU treatment and OXPHOS inhibition could be demonstrated through unbiased mathematical simulations, we performed quantitative ¹³C metabolic flux analysis (¹³C-MFA). After generating a metabolic model for isotope labeling enrichment and setting appropriate lactate efflux values, a simulation was conducted using the labeled metabolite abundance data obtained from isotope tracer analysis. The appropriate lactate efflux for quiescent HSC (PBS-treated HSC) was determined to 65 after experimenting with values from 0–100. The lactate efflux of 5-FU- or oligomycin-treated HSCs was higher than that of quiescent HSCs based on the observation that labeled glycolytic metabolite levels were particularly elevated in *in vitro* tracer analysis (see “Quantitative ¹³C-MFA with OpenMebius” under “Methods” for more information). As a result, the variation in the flux values of all

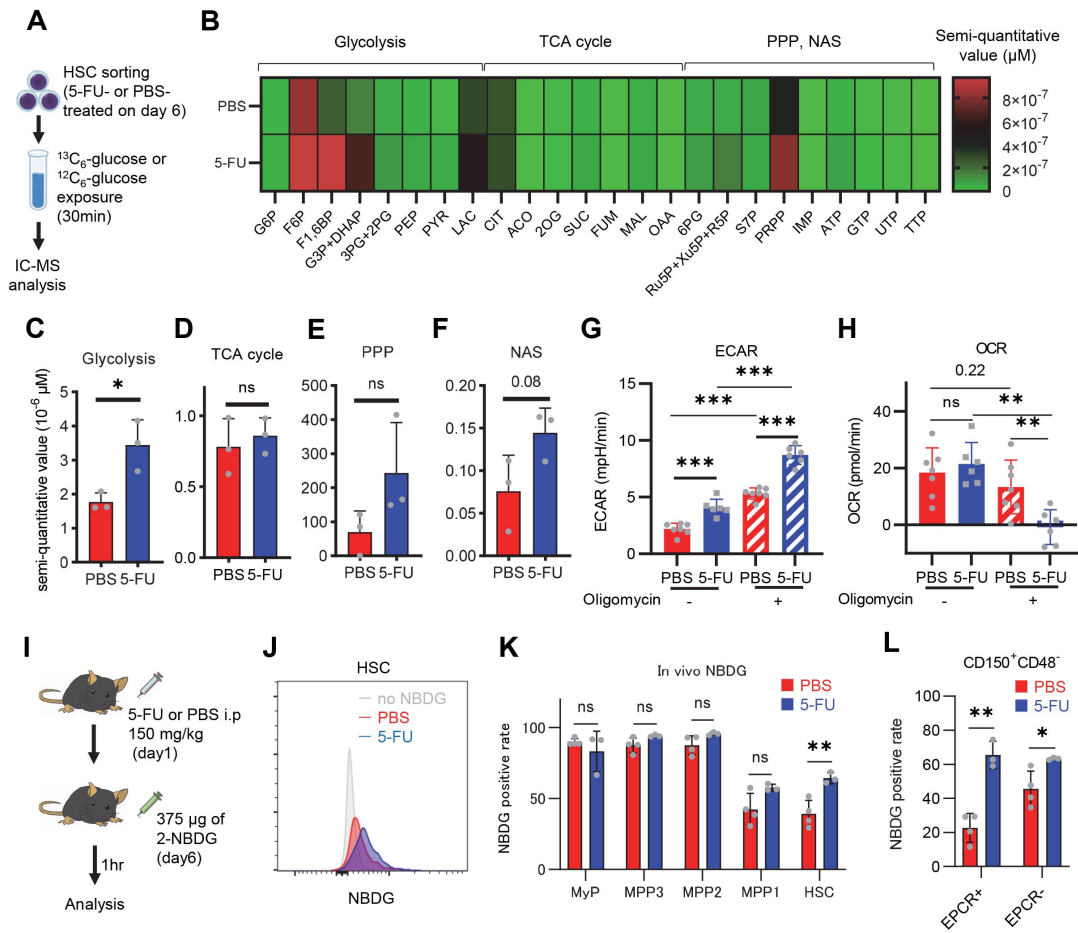


Figure 1.

HSC cell cycling increases overall glycolytic flux, but not flux into mitochondria.

(A) Experimental design used for glucose isotope tracer analysis in HSCs from 5-FU- or PBS-treated mice. **(B)** Heat map of metabolite levels in HSCs derived from mice treated with PBS or 5-FU. **(C-F)** The semi-quantitative value ($10^{-6} \mu\text{M}$) of $\text{U-}^{13}\text{C}_6$ -glucose-derived metabolites in glycolysis (C), the first round of TCA cycle (D), the PPP, and nucleotide synthesis (F) in HSCs from 5-FU- or PBS-treated mice (PBS group = 1.0); In (B)-(F), biological replicates from the PBS and 5-FU groups, obtained on three separate days, were pooled, analyzed by IC-MS, quantified based on calibration curve data for each metabolite (see “**Ion chromatography mass spectrometry (IC-MS) analysis**” section in “**Methods**” for details). **(G-H)** A Mito Stress test with the Seahorse flux analyzer on HSCs derived from mice treated with PBS or 5-FU; ECAR (G) and OCR (H) before and after oligomycin treatment. (Data were obtained from $n = 7$ technical replicates for PBS-treated HSCs and $n = 6$ for 5-FU-treated HSCs.) **(I)** Experimental schema of *in vivo* 2-NBDG analysis. **(J)** Representative histograms of 2-NBDG analysis (gray: no 2-NBDG, red: PBS group, blue: 5-FU group). **(K)** 2-NBDG positivity in each fraction; data represent four pooled biological replicates for the PBS group and three for the 5-FU group; MyP: myeloid progenitor. **(L)** EPCR expression and 2-NBDG positivity within HSC fractions. Data were extracted from each individual in (K).

Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student’s *t*-test (C-F, G-H when comparing the PBS and 5-FU groups, and K-L) or paired-samples *t*-test (G-H when comparing the conditions before and after exposure to oligomycin within the PBS/5-FU group). Abbreviations: G6P, glucose-6-phosphate; F6P, fructose-6-phosphate; F1,6BP, fructose-1,6-bisphosphate; G3P, glycerol-3-phosphate; DHAP, dihydroxyacetone phosphate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; PEP, phosphoenolpyruvate; PYR, pyruvate; LAC, lactate; Ac-CoA, acetyl-CoA; CIT, citrate; ACO, cis-aconitic acid, isocitrate; 2OG, 2-oxoglutarate; SUC, succinate; FUM, fumarate; MAL, malate; OAA, oxaloacetate; 6PG, glucose-6-phosphate; Ru5P, ribulose-5-phosphate; Xu5P, xylulose-5-phosphate; R5P, ribose-5-phosphate; S7P, sedoheptulose-7-phosphate; E4P, erythrose-4-phosphate; PRPP, phosphoribosyl pyrophosphate; IMP, inosine monophosphate; ATP, adenosine triphosphate; GTP, guanine triphosphate; UMP, uridine monophosphate; UTP, uridine triphosphate; TTP, thymidine triphosphate. See also Fig. S1-3.

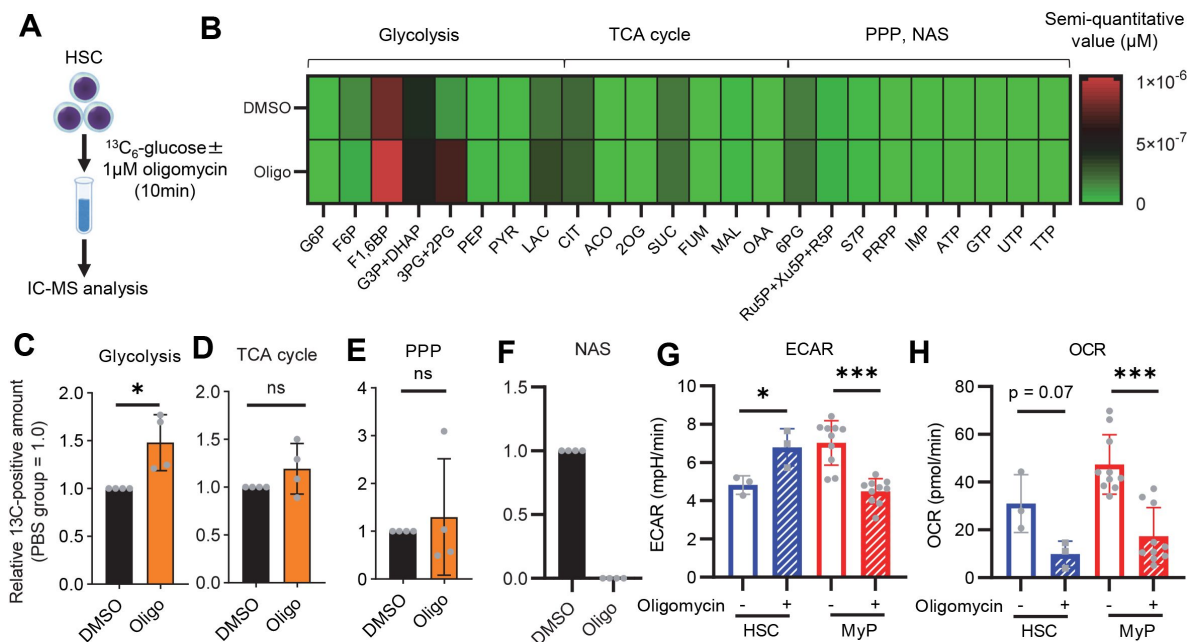


Figure 2.

OXPHOS inhibition activates compensatory glycolysis in HSCs.

(A) Experimental design used for glucose isotope tracer analysis in HSCs treated with the OXPHOS inhibitor oligomycin. **(B)** Heat map of metabolite levels detected by *in vitro* tracer analysis of $\text{U-}^{13}\text{C}_6\text{-glucose}$ in HSCs treated with DMSO or oligomycin (Oligo). **(C-F)** Relative amounts of $\text{U-}^{13}\text{C}_6\text{-glucose}$ -derived metabolites in glycolysis (C), the first round of TCA cycle (D), the PPP (E), and nucleotide synthesis (F) in DMSO-(black) or oligomycin-treated (orange) HSCs; In (B)-(F), biological replicates of the DMSO and oligomycin groups obtained on four separate days were pooled, analyzed by IC-MS, and quantified based on calibration curve data for each metabolite (see “**Ion chromatography mass spectrometry (IC-MS) analysis**” section in “**Methods**” for details). (G-H) Mito Stress test on the Seahorse flux analyzer for HSC and MyPs; ECAR (G) and OCR (H) before and after oligomycin treatment. (Data were obtained from $n = 3$ technical replicates for HSCs and $n = 10$ technical replicates for MyPs.)

Data are shown as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by paired-samples t-test (C-E and G-H).

Abbreviations: G6P, glucose-6-phosphate; F6P, fructose-6-phosphate; F1,6BP, fructose-1,6-bisphosphate; G3P, glycerol-3-phosphate; DHAP, dihydroxyacetone phosphate; 3PG, 3-phosphoglycerate; 2PG, 2-phosphoglycerate; PEP, phosphoenolpyruvate; PYR, pyruvate; LAC, lactate; Ac-CoA, acetyl-CoA; CIT, citrate; ACO, cis-aconitic acid, isocitrate; 2OG, 2-oxoglutarate; SUC, succinate; FUM, fumarate; MAL, malate; OAA, oxaloacetate; 6PG, glucose-6-phosphate; Ru5P, ribulose-5-phosphate; Xu5P, xylulose-5-phosphate; R5P, ribose-5-phosphate; S7P, sedoheptulose-7-phosphate; E4P, erythrose-4-phosphate; PRPP, phosphoribosyl pyrophosphate; IMP, inosine monophosphate; ATP, adenosine triphosphate; GTP, guanine triphosphate; UMP, uridine monophosphate; UTP, uridine triphosphate; TTP, thymidine triphosphate. See also Fig. S1–3.

enzymatic reactions calculated in HSCs after 5-FU or oligomycin treatment became smaller compared to quiescent HSCs, suggesting that HSCs strictly regulated their metabolism in response to stress (Supplemental Figure 4A-C). Unlike PBS-treated HSCs, those treated with 5-FU or oligomycin exhibited preferential glycolytic activation rather than TCA- or PPP-based metabolic strategies; the first half of the glycolytic system appeared to be the site of metabolic activation (**Figure 3A-J** [↗](#); Supplemental Figure 4D-U, Table S4). This increase in metabolic flux upstream of the glycolytic pathway was also supported by our *in vitro* tracer analysis (**Figure 1B** [↗](#) and **Figure 2B** [↗](#)), suggesting that ^{13}C -MFA was a valid metabolic simulation. Among the reactions in the first half of glycolysis, phosphorylation of fructose 6-phosphate by PFK is the irreversible and rate-limiting reaction⁵⁵ [↗](#). A detailed review of *in vitro* isotope tracer analysis results showed that the ratio of fructose 1,6-bisphosphate (F1,6BP; the product of PFK) to F6P (the substrate of PFK) was greatly elevated in HSCs during proliferation and OXPHOS inhibition (**Figure 3K-L** [↗](#)). Together with the results of quantitative ^{13}C -MFA, these findings suggested that HSCs exhibit elevated glycolytic flux relative to mitochondrial activity by increasing PFK enzyme activity under various stress conditions.

HSCs under stress exhibit activation of glycolysis-initiated TCA cycle and NAS

To investigate the long-term glucose utilization of HSCs, we performed an *in vivo* tracer analysis with $\text{U-}^{13}\text{C}_6$ glucose based on recent reports⁴⁷ [↗](#),⁵⁰ [↗](#) (Supplemental Figure 5A; see “**Preparation and storage of *in vivo* $\text{U-}^{13}\text{C}_6$ -glucose tracer samples**” under “**Methods**” for more information). In HSCs from 5-FU-treated mice, we observed increased labeling of glycolytic metabolites such as dihydroxyacetone phosphate, glycerol-3-phosphate, and phosphoenolpyruvate, as well as NAS metabolites such as inosine monophosphate and ATP, and those derived from TCA cycle such as aspartic acid and glutamate, compared to HSCs from PBS-treated mice (Supplemental Figure 5B-I, Table S5). When the amount of $\text{U-}^{13}\text{C}_6$ -glucose-derived labeled metabolites in each pathway was calculated, more glucose-derived metabolites entered TCA cycle in the 5-FU-treated group than PBS-treated group (Supplemental Figure 5J). Thus, although short-term (10–30 min) *in vitro* tracer analysis showed that HSCs exhibited more potent activation of anaerobic glycolysis than of other pathways in response to 5-FU administration, long-term (approx. 3 h) labeling by *in vivo* tracer analysis revealed that glycolysis-initiated TCA cycle and NAS flux were activated in addition to enhanced anaerobic glycolysis. Importantly, despite differences in labeling times and supplementation of $\text{U-}^{13}\text{C}_6$ glucose metabolites from non-HSCs to HSCs *in vivo*, the activation of the glycolytic system was a common finding.

PFKFB3 accelerates glycolytic ATP production during HSC cell cycling

In vitro and *in vivo* tracer analysis results collectively suggested that the activation of glycolysis catalyzed by PFK may have been the starting point for the activation of the entire HSC metabolism. To analyze the contribution of PFK to ATP metabolism in steady-state or stressed HSCs, we needed to develop an experimental system that could measure the dynamics of ATP concentrations in HSCs in a non-destructive, real-time manner. To this end, we used knock-in GO-ATeam2 mice as a FRET-based biosensor of ATP concentration (see “**Conversion of GO-ATeam2 fluorescence to ATP concentration**” under “**Methods**” for more information.). The number of bone marrow mononuclear cells (BMMNCs), as well as the frequency of HSCs ($\text{CD150}^+\text{CD48}^-\text{LSK}$) and other progenitor cells, in the bone marrow (BM) of GO-ATeam2⁺ mice were almost unchanged compared to of C57B6/J mice, except for a mild decrease in the Lin[−] fraction (Supplementary Figure 6A-C). Using BMMNCs derived from GO-ATeam2⁺ mice, we developed a method to detect changes in ATP concentration with high temporal resolution when the activity of PFK was modulated (Supplemental Figure 6D-F). To validate our methods, we measured ATP concentrations in HSCs and MyPs with or without various nutrients (see “**Time-course analysis of FRET values**” under

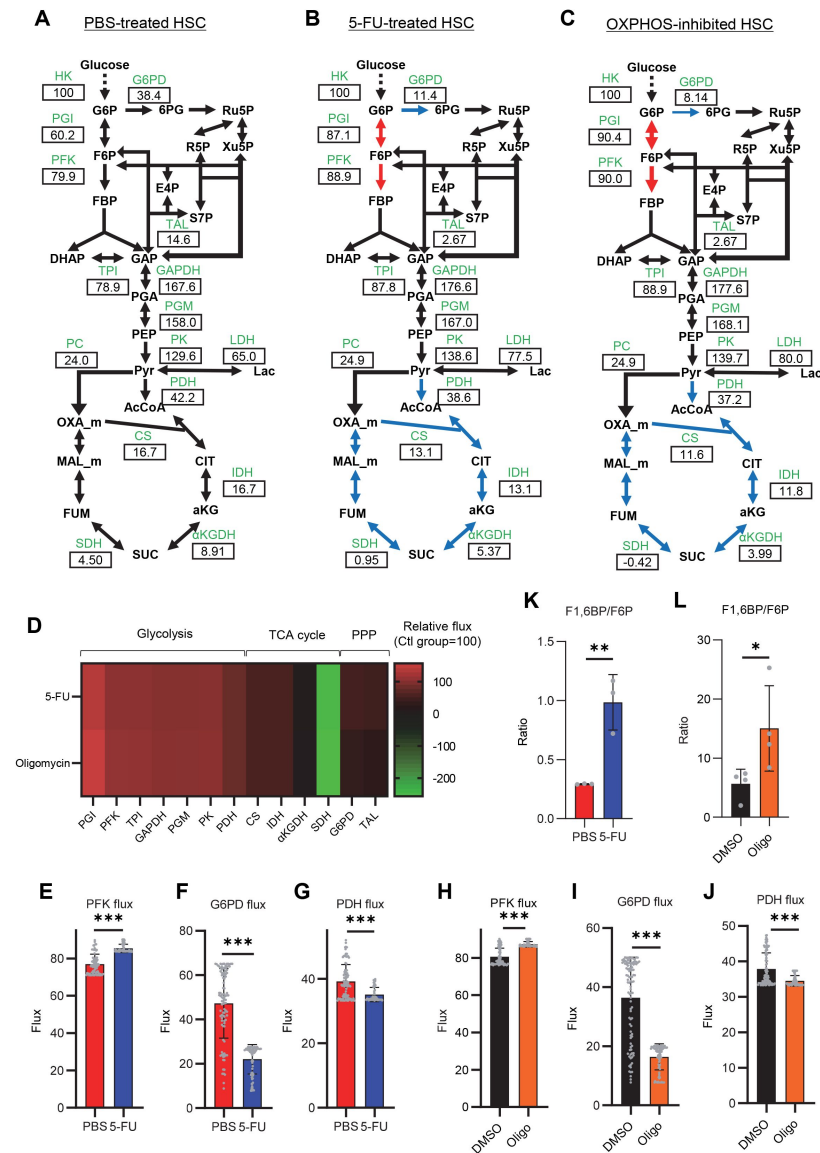


Figure 3.

Quantitative ¹³C-MFA of quiescent, proliferative, and stressed HSCs

(A–C) Overview of quantitative ¹³C-MFA of PBS-treated HSCs (A), 5-FU-treated HSCs (B), and OXPHOS-inhibited HSCs (C). The representative net flux for each reaction with glucose uptake as 100 is shown in the squares below the catalytic enzymes for each reaction listed in green letters. Red arrows indicate reactions with particularly elevated fluxes and blue arrows indicate reactions with particularly decreased fluxes. (D) Heatmap of the relative flux of each enzyme in the 5-FU or oligomycin groups compared to that in the quiescent (Ctl) HSC (The metabolic flux of each enzyme in the Ctl group was standardized as 100.). (E–J) Fluxes due to reactions with PFK (E, H), G6PD (F, I), and PDH (G, J). Fluxes of HSCs derived from mice treated with 5-FU (blue bars) or PBS (red bars) (D–F) and of HSCs treated with DMSO (black bars) or oligomycin (orange bars) (G–I) are shown. Data is obtained from 100 simulations in OpenMebius, and flux data for each enzyme is displayed (Table S4). (K–L) Ratio of fructose 1,6-bisphosphate (F1,6BP) to fructose-6-phosphate (F6P) calculated from tracer experiments shown in Figure 1B and Figure 2B. Effects of 5-FU administration (K) or mitochondrial inhibition by oligomycin (L) are summarized. Data are shown as mean ± SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student's *t*-test (E–L). Abbreviations: HK, hexokinase; PGI, glucose-6-phosphate isomerase; PFK, phosphofructokinase; TPI, triose phosphate isomerase; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; PGM, phosphoglycerate mutase; PK, pyruvate kinase; LDH, lactate dehydrogenase; PC, pyruvate carboxylase; PDH, pyruvate dehydrogenase; CS, citrate synthase; IDH, isocitrate dehydrogenase; aKGDH, α-ketoglutaric acid dehydrogenase; SDH, succinate dehydrogenase; G6PD, glucose-6-phosphate dehydrogenase; TAL, transaldolase. See also Fig. S4.

"**Methods**" for more information.). The number of bone marrow mononuclear cells (BMMNCs), as well as the frequency of HSCs (CD150⁺CD48⁺LSK) and other progenitor cells, in the bone marrow (BM) of GO-ATeam2⁺ mice were almost unchanged compared to of C57B6/J mice, except for a mild decrease in the Lin⁺ fraction (Supplementary Figure 6A-C). Using BMMNCs derived from GO-ATeam2⁺ mice, we developed a method to detect changes in ATP concentration with high temporal resolution when the activity of PFK was modulated (Supplemental Figure 6D-F). To validate our methods, we measured ATP concentrations in HSCs and MyPs with or without various nutrients (see "**Time-course analysis of FRET values**" under "**Methods**" for more information.). MyPs showed more rapid decreases in ATP concentration than HSCs, suggesting higher ATP consumption by progenitors (Supplemental Figure 6G-H). Adding glucose to the medium suppressed this decrease in MyPs; however, other metabolites (e.g., pyruvate, lactate, and fatty acids) had minimal effects, suggesting that ATP levels are glycolysis-dependent in MyPs (Supplemental Figure 6G-H), consistent with previous reports that the aerobic glycolytic enzyme M2 pyruvate kinase isoform (PKM2) is required for progenitor cell function³³.

Further, we analyzed ATP consumption and metabolic dependency of cell-cycling HSCs after 5-FU administration (**Figure 4A**). After inhibiting glycolysis using 2-deoxy-D-glucose (2-DG) with other mitochondrial substrates, 5-FU-treated HSCs showed more rapid decreases in ATP concentration than PBS-treated HSCs (**Figure 4B-C**). In contrast, OXPHOS inhibition by oligomycin without glucose or mitochondrial substrates decreased the ATP concentration to a similar extent in both 5-FU- and PBS-treated HSCs, although 5-FU-treated HSCs showed earlier ATP exhaustion (**Figure 4D-E**). These data suggest that 5-FU-treated-HSCs upregulated ATP production via glycolysis, rather than relying on mitochondria. Apoptosis assay revealed a slight increase in early apoptotic cells (annexin V⁺ propidium iodide [PI]⁺) after 2-DG treatment and a slight decrease in the number of viable cells (Annexin V⁺ PI⁻) after oligomycin treatment, both to a very limited extent (approx. 5%) compared to the degree of ATP decrease, suggesting that the decrease in ATP after 2-DG or oligomycin treatment did not simply reflect cell death (Supplementary Figure 6I). Importantly, no metabolic changes in glycolysis or OXPHOS were observed in HSCs without cell cycle progression after 5-FU administration (very early phase: day 3; late phase: day 15) (Supplementary Figure 7A-H).

PFK is allosterically activated by 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase (PFKFB). Among the four isozymes of mammalian PFKFB, PFKFB3 is the most favorable for PFK activation⁵⁶, and is the most highly expressed in HSCs (**Figure 4F**). Therefore, we investigated whether PFKFB3 contributes to glycolytic plasticity in HSCs during proliferation. When treated with the PFKFB3-specific inhibitor AZ PFKFB3 26⁵⁷, compared with HSCs from PBS-treated mice, HSCs from 5-FU-treated mice showed decreased ATP levels (**Figure 4G, I**; Supplementary Figure 7I). Although AMPK activates PFKFB3 in other contexts⁵⁸, AMPK inhibition by dorsomorphin did not alter ATP concentration in 5-FU-treated-HSCs (**Figure 4H, J**).

Finally, we investigated the nutrients that drive OXPHOS in PBS- or 5-FU-treated HSCs. Exposure of PBS- or 5-FU-treated HSCs to either etomoxir, a FAO inhibitor, or 6-diazo-5-oxo-L-norleucine (DON), a glutaminolysis inhibitor, alone or in combination, did not decrease ATP concentrations (Supplementary Figure 7J-M). Subsequent assessment of FAO activity using FAOBlue, a fluorescent probe for the FAO activity assay⁵⁹, showed no significant differences between PBS- and 5-FU-treated HSCs (Supplementary Figure 7N). Thus, neither FAO nor glutaminolysis appeared to be essential for the short-term maintenance of ATP levels in cell-cycling HSCs after 5-FU administration. Notably, the addition of glucose and a Pfkfb3 inhibitor to etomoxir rapidly reduced ATP concentrations in HSCs (Supplementary Figure 7O-P). This suggests that etomoxir may partially mimic the effects of oligomycin, indicating that OXPHOS is primarily driven by FAO, but can be compensated by Pfkfb3-accelerated glycolysis in HSCs. Conversely, exposure of HSCs to DON in combination with a Pfkfb3 inhibitor did not decrease ATP concentrations (Supplementary Figure 7O-P), suggesting that ATP production via glutaminolysis is limited in HSCs.

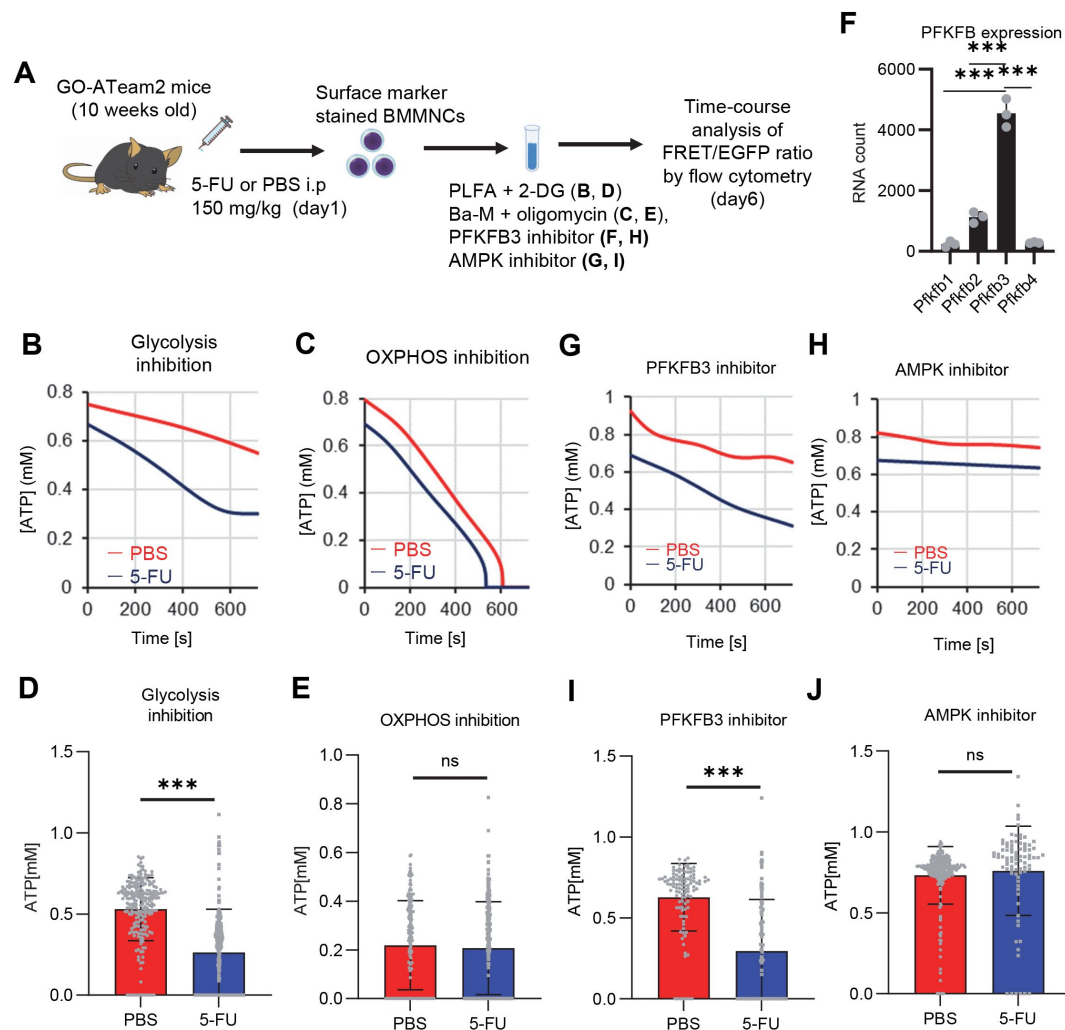


Figure 4.

Pfkfb3 activates the glycolytic system in proliferating HSCs

(A) Experimental design used to conduct real-time ATP analysis of HSCs treated with 5-FU or PBS. PLFA medium containing mitochondrial substrates (pyruvate, lactate, fatty acids, and amino acids) but no glucose, was used for experiments with 2-DG; Ba-M containing neither mitochondrial substrates nor glucose was used for experiments with oligomycin, Pfkfb3 inhibitor, or AMPK inhibitor. **(B–I)** Results of real-time ATP analysis of PBS-(red) or 5-FU-treated (blue) HSCs after treatment with 2-DG (B, D), oligomycin (C, E), PFKFB3 inhibitor (F, H), or AMPK inhibitor (G, I). Bar graphs show corrected ATP concentrations for the last 2 min (D) of (B), 6–7 min (E) of (C), or the last 1 min (H, I) of (F, G) for PFKFB3 and AMPK inhibitors, respectively. Each group represents at least 60 cells. Data are representative results of pooled samples from three biological replicates. (see “Time-course analysis of FRET values” in “Methods” for details of the correction method used to calculate ATP concentration.) **(J)** Normalized mRNA counts of PFKFB isozymes based on the RNA sequencing of HSCs. Data are presented as mean ± SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student’s *t*-test (D, E, H, and I) or a one-way ANOVA followed by Tukey’s test (J). See also Fig. S6–7.

OXPHOS inhibition accelerates glycolysis to sustain ATP levels in HSCs, but not in progenitors

To assess differences in metabolic dependence between steady-state or stressed and naturally proliferating HPCs, we altered ATP metabolism in HSCs and progenitors using 2-DG or oligomycin (**Figure 5A**). Oligomycin treatment rapidly depleted ATP in HSCs and all HPC fractions (green lines in **Figure 5B-C**; Supplemental Figure 8A-D). Treatment with 2-DG decreased ATP concentrations for a short amount of time (~12 min) in HSCs and HPCs, but ATP reduction was less evident than that induced by oligomycin (blue lines in **Figure 5B-C**; Supplemental Figure 8A-D). The ATP reduction induced by 2-DG treatment was particularly low (~15%) in HSCs, multipotent progenitor cells (MPPs), and common lymphoid progenitors (CLPs) relative to that in common myeloid progenitors (CMPs), granulocytes-macrophage progenitors (GMPs), and megakaryocyte-erythrocyte progenitors (MEPs) (**Figure 5D**).

Next, we investigated the role of glycolysis in ATP production during OXPHOS inhibition by combining oligomycin administration and glucose supplementation. ATP concentration remained more stable in HSCs treated with oligomycin and glucose than in those treated only with oligomycin. Similar results were not seen in HPCs, indicating that HSCs have the plasticity to upregulate glycolytic ATP production to meet demands (orange lines in **Figure 5B-C**; Supplemental Figure 8A-D, summarized in **Figure 5E**). Similar to oligomycin treatment, rotenone (Complex I inhibitor) and carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone (FCCP, mitochondrial uncoupler) treatments, which inhibit OXPHOS-derived ATP production, also decreased ATP concentrations in HSCs, but not when administered simultaneously with glucose (Supplementary Figure 8E-F). Furthermore, with oligomycin, HSCs, but not HPCs, maintained ATP concentrations at low glucose levels (50 mg/dL) (Supplemental Figure 8G). These analyses suggest that ATP was produced by mitochondrial OXPHOS in steady-state HSCs, and that only HSCs, but not HPCs, maintained ATP production by glycolysis when OXPHOS was compromised.

PFKFB3 accelerates glycolytic ATP production during OXPHOS inhibition

Next, to understand whether PFKFB3 contributes to ATP production in HSCs under OXPHOS inhibition, we evaluated PFKFB3 function under OXPHOS inhibition using the GO-ATeam2⁺ BMMNCs. In oligomycin-treated HSCs, PFKFB3 inhibition led to rapidly decreased ATP concentration that was not observed in HSCs not treated with oligomycin (**Figure 5F-I**). We examined the effects of HSPC metabolic regulators on ATP levels in oligomycin-treated HSCs. Inhibiting PKM2, which accelerates glycolysis in steady-state progenitors³³, significantly reduced ATP levels in oligomycin-treated HSCs (Supplemental Figure 8H, J). Inhibiting LKB1, a kinase upstream of AMPK^{60,61}, did not affect the ATP concentration in oligomycin-treated HSCs (Supplemental Figure 8I, K), whereas levels of adenosine monophosphate (AMP), which also activates AMPK, increased in oligomycin-treated but not in 5-FU-treated HSCs (Supplemental Figure 8L). This may explain differences in AMPK-dependent ATP production between proliferative HSCs and HSCs under OXPHOS inhibition.

Next, we tested the effects of PFKFB3 on ATP concentration in HPCs. Unlike HSCs, HPCs exhibited PFKFB3-dependent ATP production, even without oligomycin (Supplemental Figure 8M-Q). Therefore, ATP production in steady-state HSCs was PFKFB3-independent, and proliferative stimulation or OXPHOS inhibition plastically activated glycolytic ATP production in a PFKFB3-dependent manner to meet ATP demand.

PFKFB3 activity renders HSCs dependent on glycolysis

Next, we investigated whether PFKFB3 activity itself confers glycolytic dependence on HSCs. We retrovirally overexpressed *Pfkfb3* in HSCs and performed cell cycle analysis (**Figure 5J**). *Pfkfb3*-overexpressed HSCs increased the proportion of cells in the S/G2/M phase and decreased the

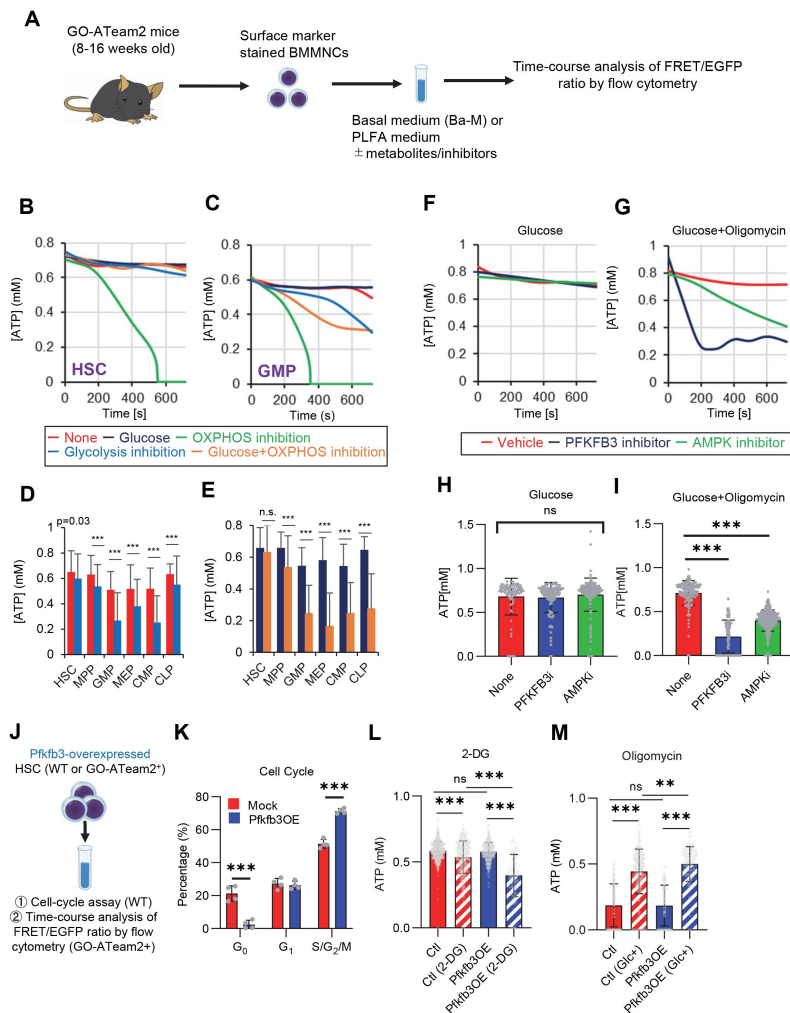


Figure 5.

PFKFB3 accelerates glycolysis in HSCs under OXPHOS inhibition in an AMPK-dependent manner

(A) Experimental design of real-time ATP analysis using GO-ATeam2 knock-in BMMNCs. Ba-M was used in experiments with oligomycin. For other experiments, PLFA medium was used. **(B-C)** Evaluation of factors affecting ATP concentration in HSCs (B) and GMPs (C) based on the GO-ATeam2 system. GO-ATeam2 knock-in BMMNCs were incubated with glucose, oligomycin, 2-DG, or glucose plus oligomycin, and the FRET/EGFP ratio was calculated. **(D)** ATP concentration in indicated stem/progenitor fractions in PLFA medium (red bars) alone or PLFA medium plus 2-DG (blue bars). ATP concentration for the last 2 min of the analysis time is shown. Data is summarized from **Figure 5B-C** and Supplemental Figure 5A-D. Each group represents at least 110 cells. Data are representative results of pooled samples from three biological replicates. **(E)** ATP concentration in indicated stem/progenitor fractions in Ba-M plus glucose (dark blue bars) or Ba-M plus glucose and oligomycin (orange bars). ATP concentration for the last 1 min of the analysis period is shown. Data is summarized from **Figure 5B-C** and Supplemental Figure 5A-D. Each group represents at least 43 cells. Data are representative results of pooled samples from three biological replicates. **(F-I)** Effects of PFKFB3 or AMPK inhibitors (PFKFB3i or AMPKi, respectively) on ATP concentration in HSCs from GO-ATeam2 mice in Ba-M plus glucose only (F) or Ba-M plus glucose and oligomycin (G). ATP concentrations for the last 1 min of the analysis period are shown in (H) and (I) for glucose only and glucose with oligomycin groups, respectively. Each group represents at least 90 cells. Data are representative results of pooled samples from three biological replicates. **(J)** Experimental schema for cell cycle assay and real-time ATP concentration analysis after overexpression of *Pfkfb3*. **(K)** Cell cycle status of *Pfkfb3*-overexpressing (*Pfkfb3*OE) and mock-transduced HSCs. **(L-M)** Effects of inhibitors on ATP concentration in *Pfkfb3*-overexpressing GO-ATeam2⁺ HSCs. Cells were exposed to vehicle or 2-DG (L), oligomycin in the presence or absence of glucose 12.5 mg/dL (M), and ATP concentrations for the last 2 min (L) or 1 min (M) of the analysis period were calculated. Data are representative results of pooled samples from three biological replicates. Data are presented as mean ± SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student's *t*-test (D, E, and K) or one-way ANOVA followed by Tukey's test (H, I, L, and M). See also Fig. S8.

number of G₀ cells compared to *mock*-overexpressed HSCs (**Figure 5K**). Next, we retrovirally overexpressed *Pfkfb3* in GO-ATeam2⁺ HSCs and performed real-time ATP measurement (**Figure 5J**). *Pfkfb3*-overexpressing GO-ATeam2⁺ HSCs did not show changes in ATP concentrations relative to those in *mock*-transduced cells (**Figure 5L**; Supplemental Figure 8R). Upon 2-DG treatment, *Pfkfb3*-overexpressing HSCs showed a greater decrease in ATP concentration than *mock*-transduced HSCs did (**Figure 5L**; Supplemental Figure 8S). However, oligomycin treatment of both *mock*-transduced and *Pfkfb3*-overexpressing HSCs decreased ATP concentration to comparable levels (**Figure 5M**; Supplemental Figure 8T). Notably, *Pfkfb3*-overexpressing HSCs recovered ATP levels more effectively under low glucose conditions (12.5 mg/dL) than did *mock*-transduced HSCs (**Figure 5M**; Supplemental Figure 8U). These data suggest that PFKFB3 directly conferred glycolytic dependence onto HSCs by modulating the cell cycle and increasing their ATP-generating capacity via glycolysis under metabolic stress.

PFKFB3 methylation by PRMT1 supports ATP production by cell-cycling HSCs

Next, we investigated how 5-FU-treated-HSCs regulate PFKFB3 independently of AMPK (**Figure 4G-J**). PFKFB3 activity is regulated at multiple levels⁶², and PFKFB3 transcript and protein levels in HSCs remained unchanged during 5-FU-induced cell cycling (**Figure 6A-B**). Phosphorylation can also regulate PFKFB3 activity^{58,63,64}; however, we observed no change in PFKFB3 phosphorylation in 5-FU-treated-HSCs (**Figure 6C**). Upon oligomycin exposure, PFKFB3 was phosphorylated by AMPK in the HSCs (**Figure 6D**). PFKFB3 is also methylated, and its activity is upregulated by protein arginine methyltransferase 1 (PRMT1)⁶⁵. We observed that *Prmt1* expression increased in 5-FU-treated-HSCs relative to that in PBS-treated-HSCs (**Figure 6E**). Furthermore, PFKFB3 methylation was significantly induced in 5-FU-treated-HSCs than in PBS-treated-HSCs (**Figure 6F**). Treatment of HSCs with a PRMT1 inhibitor decreased PFKFB3 methylation (**Figure 6G**), suggesting that PRMT1 catalyzed PFKFB3 methylation. In contrast, the number of transcripts regulated by PRMT1 decreased or was unchanged (Supplementary Figure 9), suggesting that the transcriptional regulatory function of PRMT1 is limited. To investigate whether glycolytic activity in HSCs was regulated by m-PFKFB3, mice treated with PBS or 5-FU were injected with 2-NBDG, and m-PFKFB3 levels in HSCs with high and low 2-NBDG uptake were quantified. Regardless of PBS or 5-FU treatment, HSCs with high 2-NBDG uptake exhibited higher m-PFKFB3 levels than those with low uptake (**Figure 6H**), suggesting that m-PFKFB3 regulated the activity of the glycolytic system in HSCs.

Further, we analyzed the potential effects of PRMT1 inhibition on ATP concentration in GO-ATeam2⁺ HSCs. Treatment with the PRMT1 inhibitor significantly decreased ATP levels in 5-FU-treated-HSCs than in PBS-treated-HSCs (**Figure 6I**). In contrast, the retroviral overexpression of *Pfkfb3* in GO-ATeam2⁺ HSCs abolished the effect of the PRMT1 inhibitor on ATP reduction (**Figure 6J**). These findings indicated that ATP levels in 5-FU-treated-HSCs were supported by PRMT1 methylation-mediated PFKFB3 activation.

PFKFB3 contributes to HSPC pool expansion and stress hematopoiesis maintenance

Finally, we analyzed PFKFB3 function in HSCs during hematopoiesis. We cultured HSCs with a PFKFB3 inhibitor *in vitro* under quiescence-maintaining or proliferative conditions (Supplementary Figure 10A). Cell count in HSC-derived colonies decreased following treatment with a PFKFB3 inhibitor under proliferative, but not quiescence-maintaining, conditions (Supplementary Figure 10B). We also knocked out *Pfkfb3* in HSCs using the less toxic, vector-free CRISPR-Cas9 system and cultured the cells under quiescence-maintaining or proliferative conditions (Supplementary Figure 10A) based on recent reports by Shiroshita et al.⁶⁶. Again, cell numbers in *Pfkfb3*-knockout (KO) HSC-derived colonies decreased only in proliferative cultures when compared to control cultures (*Rosa26*-KO HSCs) (Supplementary Figure 10C, E, F). We

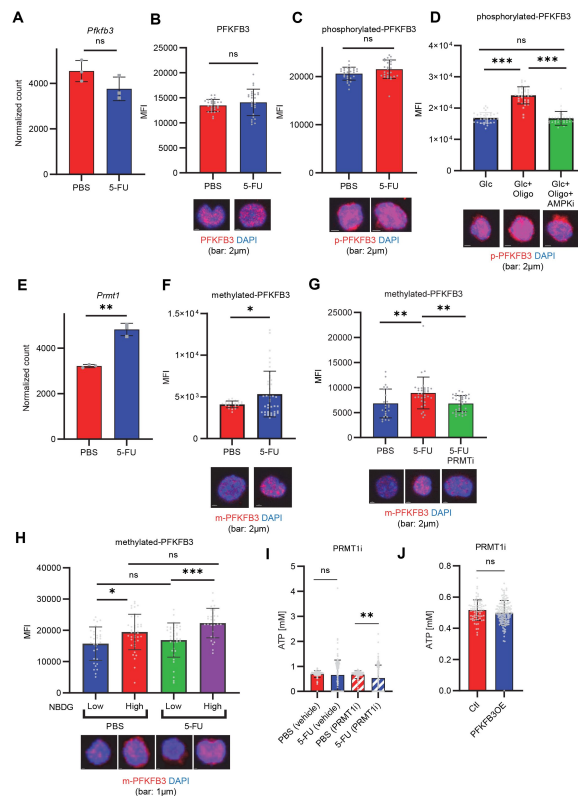


Figure 6.

PFKFB3 methylation by PRMT1 enables ATP production by cell-cycling HSCs.

(A) Normalized *Pfkfb3* mRNA counts based on RNA sequencing of PBS-treated (red) or 5-FU-treated (blue) HSCs. Data are representative results of pooled samples from three biological replicates. Data were extracted from the same pooled samples as in **Figure 4J** and Figure S9. **(B)** Quantification of mean fluorescent intensity (MFI) of PFKFB3 protein in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of PFKFB3 in each group. $n = 26$ – 27 single HSCs for each group. The data are representative results from two independent experiments. **(C)** Quantification of MFI of phosphorylated-PFKFB3 (p-PFKFB3) protein in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of p-PFKFB3 in each group. $n = 27$ single HSCs for each group. The data are representative results from two independent experiments. **(D)** Quantification of mean fluorescence intensity (MFI) of p-PFKFB3 in HSCs treated with glucose (200mg/dL); glucose plus oligomycin (1 μ M); and glucose, oligomycin, and dorsomorphin (100 μ M) for 5 min. The lower part of the graph shows representative images of immunocytochemistry of p-PFKFB3 in each group. $n = 32$ – 36 for each group. The data are representative results from two independent experiments. **(E)** Normalized *Prmt1* mRNA counts based on RNA sequencing of PBS-treated (red) or 5-FU-treated (blue) HSCs. Data are representative results of pooled samples from three biological replicates. **(F)** MFI quantification of methylated-PFKFB3 (m-PFKFB3) in PBS- or 5-FU-treated HSCs. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. $n = 23$ – 41 for each group. The data are representative results from three independent experiments. **(G)** Quantification of MFI of m-PFKFB3 in PBS- or 5-FU-treated HSCs or 5-FU-treated HSCs after 15 min treatment with a PRMT1 inhibitor (90 μ g/mL GSK3368715); $n = 25$ – 35 single HSCs for each group. The lower part of the graph shows representative images showing immunocytochemistry of m-PFKFB3. Data represent a single experiment. **(H)** Quantitation of m-PFKFB3 in NBDG-positive or -negative HSCs in mice treated with PBS or 5-FU. The lower part of the graph shows representative images of immunocytochemistry of m-PFKFB3 in each group. $n = 28$ – 41 for each group. The data are representative results from two independent experiments. **(I)** Corrected ATP levels in PBS- (red) or 5-FU-treated (blue) HSCs 15 min after treatment with vehicle or a PRMT1 inhibitor (90 μ g/mL GSK3368715). Each group represents at least 101 cells. Data are representative results of pooled samples of two biological replicates. (See “Time-course analysis of FRET values” in “Methods” for details of the correction method used to calculate ATP concentration.) **(J)** ATP concentration in mock-transduced (Ctl) or PFKFB3-overexpressed (OE) HSCs after treatment with the PRMT1 inhibitor (90 μ g/mL GSK3368715). ATP concentration for the last 1 min of the analysis period is shown. Data are presented as mean $\pm$ SD. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$ as determined by Student’s *t*-test (A–C, E–F, and I–J) or one-way ANOVA followed by Tukey’s test (D, G, and H). See also Fig. S9.

retrovirally overexpressed *Pfkfb3* in HSCs and cultured them under quiescence maintenance or proliferative conditions (Supplementary Figure 10A). *Pfkfb3*-overexpressing HSC colonies showed increased cell count compared to that of *mock*-transduced cells, but only under proliferative conditions (Supplementary Figure 10D).

To assess PFKFB3 function in HSCs *in vivo*, we transplanted *Pfkfb3*-KO HSCs (Ly5.2⁺) or wild type (WT) control HSCs into lethally irradiated recipients (Ly5.1⁺) as well as Ly5.1⁺ competitor cells (Figure 7A), and the behavior of *Pfkfb3*-KO cells was evaluated by Sanger sequencing of peripheral blood (PB) cells⁶⁶. In the KO group, donor-derived chimerism in PB cells decreased relative to that in the WT control group during the early phase (1 month post-transplant) but recovered thereafter (Figure 7B). Next, we retrovirally transduced Ly5.2⁺ HSCs with *Pfkfb3* S461E (*Pfkfb3*CA), a constitutively active PFKFB3 mutant, and transplanted them into lethally irradiated recipients (Ly5.2⁺), along with Ly5.1⁺ competitor cells (Figure 7A, Supplementary Figure 10G). Donor chimerism during the early post-transplant period in the *Pfkfb3*CA-overexpressing group was significantly higher than that in the *mock*-transduced group (Figure 7C). These findings suggest that PFKFB3 may play a role in the differentiation and proliferation of HSCs. Therefore, we compared the contribution of PFKFB3 to HSPC function at steady state and after myeloproliferative stimulation. *Pfkfb3*- or *Rosa*-KO HSPCs were transplanted into recipients (Ly5.1⁺). After 2 months, recipients received 5-FU intraperitoneally, and the dynamics of *Pfkfb3*- or *Rosa*-KO cell abundance in PB was assessed (Figure 7D). In PB cells prior to 5-FU administration, *Pfkfb3*- or *Rosa*-KO HSPC-derived blood cells were almost equally present, suggesting a limited involvement of PFKFB3 in steady-state blood cell production (Figure 7E). However, after 5-FU administration, *Pfkfb3*-KO HSPC-derived blood cell abundance was reduced compared to that in the *Rosa* group (Figure 7E). This change occurred on day 6 after 5-FU administration (day 1), when the cell cycle of HSCs was activated (Supplementary Figure 1D), supporting the idea that PFKFB3 contributes to HSC proliferation and differentiation into HSPCs.

To investigate the mechanisms underlying the short-term effects of PFKFB3 on hematopoiesis after bone marrow transplantation (BMT), we evaluated cell cycle and apoptosis of *Pfkfb3*-KO or -overexpressing HSPCs on day 2 after BMT (Figure 7F). Cell cycle was analyzed by Ki67/Hoechst33342 staining and *in vivo* BrdU labeling⁵⁰, which showed that cell cycle progression was suppressed in *Pfkfb3*-KO HSPCs (Figure 7G-J). In contrast, *Pfkfb3*-KO cells did not show increased apoptotic rates or decreased homing efficiency after BMT (Figure 7K; Supplementary Figure 10H). Furthermore, we examined the cell cycle of HSPCs overexpressing *Pfkfb3*CA on day 2 after BMT (Figure 7L) and found that *Pfkfb3*CA-overexpressing HSPCs showed accelerated cell cycle compared to *mock*-overexpressing HSPCs (Figure 7M-N). These data suggest that PFKFB3 contributes to HSC proliferation and differentiates cell production in *in vitro* and *in vivo* proliferative environments (cytokine stimulation and transplantation).

Discussion

In this study, by combining metabolomic tracing of U-¹³C₆-labeled glucose and ¹³C-MFA, we quantitatively identified the metabolic programs used by HSCs during steady-state, cell-cycling, and OXPHOS inhibition. Under proliferative stress, HSCs uniformly shift from mitochondrial respiration to glycolytic ATP production and PPP activation, which represent hallmarks of cell-cycling mammalian cells⁶⁷. Previous reports have emphasized the importance of glycolysis in maintaining HSC quiescence, but have primarily analyzed HSCs in transplant assays, wherein HSCs must enter the cell cycle^{32,68}. Prior analysis of repopulation capacity, which is positively correlated with enhanced glycolysis, may have overestimated glycolytic ATP production and overlooked mitochondrial ATP production during native hematopoiesis. In fact, some studies have suggested that OXPHOS activity is important for HSC maintenance and function²¹.

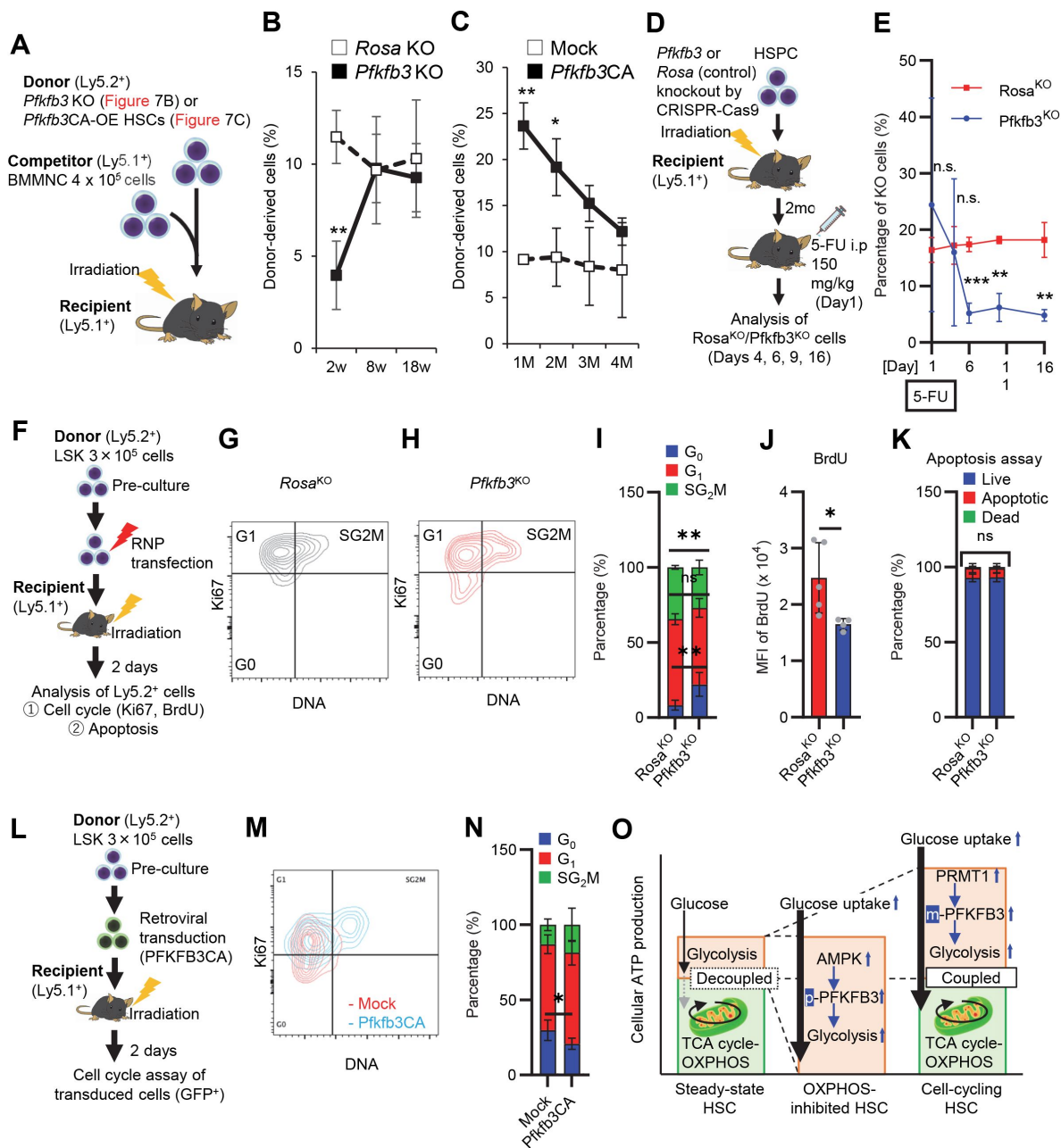


Figure 7.

PFKFB3 maintains HSC function under proliferative stress.

(A-C) Transplant analysis of *Pfkfb3*-KO or *Pfkfb3*CA-overexpressing HSCs. Experimental design (B). PB chimerism of donor-derived cells at 4 months post-transplant. *Pfkfb3*-KO group, *n* = 6; *Rosa*-KO group, *n* = 4; (B) *Pfkfb3* group, *n* = 5; pMY-IRES-GFP group, *n* = 4. (C) The data are representative results from two independent experiments. (D-E) 5-FU administration after bone marrow reconstruction with *Pfkfb3*- or *Rosa*-KO HSCs. Experimental schema (D). Behavior of the *Pfkfb3*- or *Rosa*-KO cells in PB after 5-FU administration (E). *n* = 5 for each group. (F-K) Cell cycle analysis and apoptosis assay of *Pfkfb3*- or *Rosa*-KO HSCs on day 2 post-BMT. Experimental schema (F). Representative plots of Ki67/Hoechst33432 staining of *Rosa*-KO (G) or *Pfkfb3*-KO (H) HSCs and summary of analysis (I); summary of *in vivo* BrdU labeling assay (J). Apoptosis assay results (K). *n* = 4-5 biological replicates for each group. (L-N) Cell cycle analysis of *Pfkfb3*CA or Mock-overexpressing HSCs on day 2 after BMT. Experimental Schema (L). Representative plot of Ki67/Hoechst33432 staining for both groups (M) and summary of analysis (N). *n* = 5 biological replicates for each group. (O) Models showing ATP production and regulation in quiescent, OXPHOS-inhibited, and cell-cycling HSCs. Note that the GO-ATeam2 system identified plastic acceleration of glycolysis by PFKFB3 in response to different types of stress maintains ATP levels. Data are presented as mean ± SD. * *p* ≤ 0.05, ** *p* ≤ 0.01, *** *p* ≤ 0.001 as determined by Student's *t*-test (B, C, E, H, J, K, and N). See also Fig. S10.

Our method was based on recently reported quantitative metabolic analysis techniques for very small numbers of cells^{23,46–50}, such as HSCs, and expands our knowledge of HSC metabolism during stress hematopoiesis. In our study, 5-FU administration in mice transiently decreased ATP concentration in HSCs in parallel with cell cycle progression, suggesting that HSC differentiation and cell cycle progression are closely related to intracellular metabolism and can be monitored by measuring ATP concentration. We mainly analyzed a mixture of EPCR⁺ and EPCR[−] HSCs, and we believe that the observed cell cycle progression and promotion of glycolysis in both EPCR⁺ and EPCR[−] HSCs support the validity of our claims (**Figure 1L**, Supplementary Figure 1G–K). According to ¹³C-MFA enzymatic reaction flux of PFK in 5-FU-treated HSCs indicated a relative increase of approximately 10%. However, the flux value obtained by ¹³C-MFA was calculated with glucose uptake as 100. Thus, when combined with the overall increase in the glycolytic pool demonstrated by *in vitro* isotopic glucose tracer analysis and *in vivo* NBDG analysis, rapid acceleration of glycolysis becomes evident throughout the HSCs, including subpopulations that were less responsive to stress^{43–45}. These findings are consistent with reports suggesting that HSCs have relatively low biosynthetic activity^{69,70} that is rapidly activated in response to cell proliferation stimuli^{40,71}. Notably, we found that HSCs could accelerate glycolytic ATP production to fully compensate for mitochondrial ATP production under OXPHOS inhibition, a phenomenon that is difficult to identify without real-time ATP analysis. Thus, HSCs exposed to acute stresses choose to change the efficiency of glucose utilization (accelerated glycolytic ATP production) rather than other energy sources. *In vivo*, a completely glucose-deficient environment is improbable. Therefore, even under conditions such as hypoxia, where OXPHOS is inhibited, it is conceivable that glycolysis is accelerated to maintain ATP concentrations. Glucose tracer analysis showed NAS suppression under OXPHOS inhibition, leading to glycolysis without cell proliferation (**Figure 2C–F**; Supplementary Figure 1N). This suppression can be attributed to several factors: phosphates derived from ATP are added to nucleotide mono-/di-phosphates during NAS; the primary source of ATP production, OXPHOS, is impaired; and the presence of enzymes, such as dihydroorotate dehydrogenase, which are conjugated with OXPHOS⁷². Such multifactorial effects raise new questions about the relationship between OXPHOS and nucleotide synthesis. On the other hand, we observed that ATP production in steady-state or cell-cycling HSCs and in naturally proliferating HPCs depended more on mitochondrial OXPHOS than on glycolysis; inhibiting glycolysis in steady-state HSCs resulted in only mild ATP decreases, suggesting that OXPHOS is still the major source of ATP production even in a medium saturated with hypoxia mimicking the BM environment. The p50 value of mitochondria (the partial pressure of oxygen at which respiration is half maximal) is less than 0.1 kPa, corresponding to an oxygen concentration of less than 0.1% under atmospheric pressure⁷³, suggesting that even under hypoxic conditions, OXPHOS can maintain some level of activity. Because FAO and the mitochondrial respiratory chain are necessary for HSC self-renewal and quiescence^{21,37,39,54}, fatty acids may support mitochondrial ATP production independently of fluxes from glycolysis. FAO and glutaminolysis were not immediately essential for ATP production in HSCs. Given reports on the long-term necessity of FAO and glutaminolysis for HSC maintenance^{39,74}, ATP concentrations could be maintained in the short term by compensatory pathways. Furthermore, although glycolysis and TCA cycle are decoupled in steady-state HSCs, in response to cell cycle progression, anaerobic glycolytic metabolism in HSCs is enhanced (**Figure 1**) and fluxes to TCA cycle and PPP from the glycolytic system are also promoted (Supplementary Figure 5). The degree of glycolysis and TCA cycle coupling observed by *in vitro* and *in vivo* tracer analysis differed, likely due to differences in labeling time (10–30 min *in vitro* and 3 h *in vivo*). In particular, *in vivo* tracer analysis allows all cells to be capable of metabolizing U-¹³C₆-glucose and providing its metabolites to HSCs, and there is a significant amount of time, approximately 120–180 minutes, after glucose labeling to purify HSCs. Metabolic reactions will continue during this time and subsequent processing on ice, which may increase the influx of labeled carbon into the TCA cycle. This complex dynamic in the *in vivo* tracer analysis makes it difficult to determine whether the labeled carbon influx is the result of direct influx from glycolysis or the re-uptake of metabolites by HSCs that have been processed by other cells. This is in contrast to *in vitro* analysis where such extended metabolic processing does not occur. Furthermore, despite an increased carbon influx into the TCA cycle *in vivo*, ATP

production from mitochondria does not show a corresponding increase after 5-FU treatment, as shown by the GO-ATeam2 analysis shown in **Figure 4C**. Despite these technical differences, an essential common finding from both *in vivo* and *in vitro* analyses is the activation of glycolysis and nucleotide synthesis (NAS) in 5-FU-treated HSCs, highlighting critical metabolic changes in response to treatment. Moreover, these data provide direct evidence that glycolysis and TCA cycle become functionally uncoupled in quiescent HSCs^{32,38}. Our findings are also consistent with previous reports of OXPHOS activation associated with HSC proliferation^{32,36,75,76}. In other words, HSCs exhibit an increased proportion of anaerobic glycolysis-derived ATP by PFKFB3 upon proliferation and OXPHOS inhibition; furthermore, the glycolytic system is the starting point of metabolic activation and is indispensable for the overall enhancement of HSC metabolism (**Figure 7H**).

HPCs and leukemic cells accelerate glycolytic ATP production using PKM2 for differentiation and transformation, respectively³³; however, we demonstrated that glycolytic acceleration does not fully compensate for mitochondrial ATP production in HPCs. Mechanistically, PFKFB3 increased glycolytic activity in HSCs to maintain ATP concentrations during proliferation and OXPHOS inhibition. Furthermore, inhibition of PFKFB3 in addition to OXPHOS does not result in a complete loss of ATP in HSCs, suggesting the robustness of HSC metabolism (**Figure 5G**). Under steady-state conditions, naturally proliferating HPCs rely on PFKFB3 for ATP production, whereas HSCs do not. This may explain the reduction of ECAR after oligomycin treatment in MyPs as shown by the Mito stress test (**Figure 2G**). In other words, while PFKFB3-dependent active glycolysis and mitochondria must always be coupled in MyPs, this is not necessarily the case in HSCs, even after 5-FU treatment (**Figure 1G**). Therefore, we can infer that quiescent HSCs at steady state can produce ATP via PFKFB3 activation in response to stress, enabling additional ATP generation. Furthermore, overexpression of *Pfkfb3* in HSCs increased glycolytic dependency, suggesting that PFKFB3 itself can modulate metabolic dependency in HSCs. Changes in glycolytic dependency in HSCs overexpressing *Pfkfb3* may seem small (0.06–0.13 mM) (**Figure 5L, M**). However, it is noteworthy that the rate of the reaction catalyzed by PFK varies greatly within a very narrow range of ATP concentrations, less than 1 mM. Webb et al. analyzed the factors controlling PFK activity and reported that the reaction rate of PFK varies by approximately 40% in the 0.3–1 mM ATP concentration range⁷⁷. The reason that differences in glycolytic dependence could be detected in cells overexpressing *Pfkfb3* may be that the ATP concentration at the time of analysis was approximately 0.5–0.6 mM, which is within the range where a small change in ATP concentration can dynamically alter PFK activity.

PFKFB3 supports hematopoiesis in contexts that require robust HSPC proliferation *in vitro* and *in vivo*. We showed that the positive or negative effect of *Pfkfb3* overexpression or KO on differentiated blood cell production is gradually lost after BMT. This is because HSPCs require PFKFB3 for cell cycle progression during stress hematopoiesis in the early phase after BMT (**Figure 7F–J**, and **L–N**). However, even during stress hematopoiesis, PFKFB3 is not involved in cell death or homing efficiency (**Figure 7K**; Supplementary Figure 10H) and appears to contribute primarily to the regulation of transient HSPC proliferation in the BM cavity. HSCs no longer require PFKFB3 for a certain period of time after BMT, probably because they regain a quiescent state. This is consistent with the fact that inhibition of PFKFB3 in quiescent HSCs does not reduce the ATP concentration (**Figure 5F, H**), suggesting that the activity of PFKFB3 is plastically modified. HSC metabolic plasticity is also illustrated by the mode of PFKFB3 activation, differing depending on stress type. During proliferative stress, PRMT1 methylates PFKFB3 in the HSCs to promote glycolytic ATP production, a modification that increases its activity⁶⁵. PRMT1 is required for stress hematopoiesis⁷⁸, but its downstream targets in HSCs remain unclear. Our results strongly suggest that PRMT1 targets PFKFB3 to stimulate glycolysis in HSCs. In contrast, under OXPHOS inhibition, PFKFB3 phosphorylation by AMPK is induced—another modification that also upregulates its activity. These two PFKFB3 protein modifications allow for flexible regulation of ATP production by glycolysis, even under simultaneous and different stresses. In fact, the constitutively active S461E PFKFB3 mutant, designed to mimic phosphorylation in response to

OXPPOS inhibition, enhanced HSC reconstitution capacity after transplantation, suggesting that even if PFKFB3 is activated by one stress (in this case, proliferative), it has the activation capacity to respond to a different stress (i.e., mitochondrial). Therefore, the functions of phosphorylated and methylated forms of PFKFB3 are to some extent interchangeable, and either modification can be used to handle diverse stresses. In summary, we found that HSCs exhibit a highly dynamic range of glycolytic flux. Our study highlights glycolysis as a pivotal source of energy production in stressed HSCs, and indicates that OXPPOS, although an important source of ATP, can be uncoupled from glycolysis in steady-state HSCs without compromising ATP levels. Because multiple PFKFB3 modifications safeguard HSCs against different stresses by accelerating glycolysis, interventions targeting these might effectively induce or manage stress hematopoiesis. This study provides a platform for comprehensive and quantitative real-time analysis of ATP concentration and its dynamics in HSPCs. Our approach allows for analysis of metabolic programs in rare cells and detection of various metabolic activities within a diverse cell population, making it applicable to the analysis of various tissue systems in normal and diseased states.

Limitations of the study

In this study, 5-FU-treated HSCs were analyzed as cell-cycling HSCs, but if more sensitive and time-saving glucose tracer analysis methods (especially after *in vivo* labeling with isotopic glucose) are developed, it may be possible to prospectively differentiate and quantitatively analyze HSC metabolism based on the cell surface antigens and cell cycle status. Although our assay uses media that mimic the BM environment, in the near future, *in vivo* GO-ATeam2 analysis will allow us to measure ATP concentrations in physiologically hypoxic BM.

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Conflict of interest

The authors declare no competing financial interests.

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Review #1 (Public Review)

Watanuki et al used metabolomic tracing strategies of U-13C6-labeled glucose and 13C-MFA to quantitatively identify the metabolic programs of HSCs during steady-state, cell-cycling, and OXPHOS inhibition. They found that 5-FU administration in mice increased anaerobic glycolytic flux and decreased ATP concentration in HSCs, suggesting that HSC differentiation and cell cycle progression are closely related to intracellular metabolism and can be monitored by measuring ATP concentration. Using the GO-ATeam2 system to analyze ATP levels in single hematopoietic cells, they found that PFKFB3 can accelerate glycolytic ATP production during HSC cell cycling by activating the rate-limiting enzyme PFK of glycolysis. Additionally, by using Pfkfb3 knockout or overexpressing strategies and conducting experiments with cytokine stimulation or transplantation stress, they found that PFKFB3 governs cell cycle progression and promotes the production of differentiated cells from HSCs in proliferative environments by activating glycolysis. Overall, in their study, Watanuki et al combined metabolomic tracing to quantitatively identify metabolic programs of HSCs and found that PFKFB3 confers glycolytic dependence onto HSCs to help coordinate their response to stress.

- <https://doi.org/10.7554/eLife.87674.2.sa1>

Review #2 (Public Review)

In the manuscript Watanuki et al. define the metabolic profile of HSCs in stress/proliferative (myelosuppression with 5-FU), and mitochondrial inhibition and homeostatic conditions. Their conclusions are that during proliferation HSCs rely more on glycolysis (as other cell types) while HSCs in homeostatic conditions are mostly dependent on mitochondrial metabolism. Mitochondrial inhibition is used to demonstrate that blocking mitochondrial metabolism results in similar features of proliferative conditions.

The authors used state-of-the-art technologies that allow metabolic readout in a limited number of cells like rare HSCs. These applications could be of help in the field since one of the major issues in studying HSCs metabolism is the limited sensitivity of the "standard" assays, which make them not suitable for HSC studies.

- <https://doi.org/10.7554/eLife.87674.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1:

Watanuki et al used metabolomic tracing strategies of U-13C6-labeled glucose and 13C-MFA to quantitatively identify the metabolic programs of HSCs during steady-state, cell-cycling, and OXPHOS inhibition. They found that 5-FU administration in mice increased anaerobic glycolytic flux and decreased ATP concentration in HSCs, suggesting that HSC differentiation and cell cycle progression are closely related to intracellular metabolism and can be monitored by measuring ATP concentration. Using the GO-ATeam2 system to analyze ATP levels in single hematopoietic cells, they found that PFKFB3 can accelerate glycolytic ATP production during HSC cell cycling by activating the rate-limiting enzyme PFK of glycolysis. Additionally, by using Pfkfb3 knockout or overexpressing strategies and conducting experiments with cytokine stimulation or transplantation stress, they found that PFKFB3 governs cell cycle progression and promotes the production of differentiated cells from HSCs in proliferative environments by activating glycolysis. Overall, in their study, Watanuki et al combined metabolomic tracing to quantitatively identify metabolic programs of HSCs and found that PFKFB3 confers glycolytic dependence onto HSCs to help coordinate their response to stress. Even so, several important questions need to be addressed as below:

We sincerely appreciate the constructive feedback from the reviewer. Additional experiments and textual improvements have been made to the manuscript based on your valuable suggestions. In particular, the major revisions are as follows: First, we investigated the extent to which other metabolites, not limited to the glycolytic system, affect metabolism in HSCs after 5-FU treatment. Second, the extent to which PFKFB3 contributes to the expansion of the HSPC pool in the bone marrow was adjusted to make the description more accurate based on the data. Finally, we overexpressed PFKFB3 in HSCs derived from GO-ATeam2 mice and confirmed that PRMT1 inhibition did not reduce the ATP concentration. We believe that the reviewer's valuable comments have further deepened our knowledge of the significance of glycolytic activation by PFKFB3 that we have demonstrated. Our response to the "Recommendations for Authors" is listed first, followed by our responses to all "Public Review" comments as follows:

(Recommendations For The Authors):

1. The methods used in key experiments should be described in more detail. For example, in the section on 'Conversion of GO-ATeam2 fluorescence to ATP concentration', the knock-in strategy for GO-ATeam2 should be described, as well as U-13C6 -glucose tracer assays.

As per your recommendation, we have described the key experimental method in more detail in the revised manuscript: the GO-ATeam2 knock-in method was reported by Yamamoto et al. 1. Briefly, they used a CAG promoter-based knock-in strategy targeting the Rosa26 locus to generate GO-ATeam2 knock-in mice. A description of the method has been added to Methods and the reference has been added to the citation.

For the U-13C6-glucose tracer analysis, the following points were added to describe the details of the analysis: First, a note was added that the number of cells used for the in vitro tracer analysis was the number of cells used for each sample. Second, we added the solution from which the cells were collected by sorting. We added that the incubation was performed under 1% O₂ and 5% CO₂.

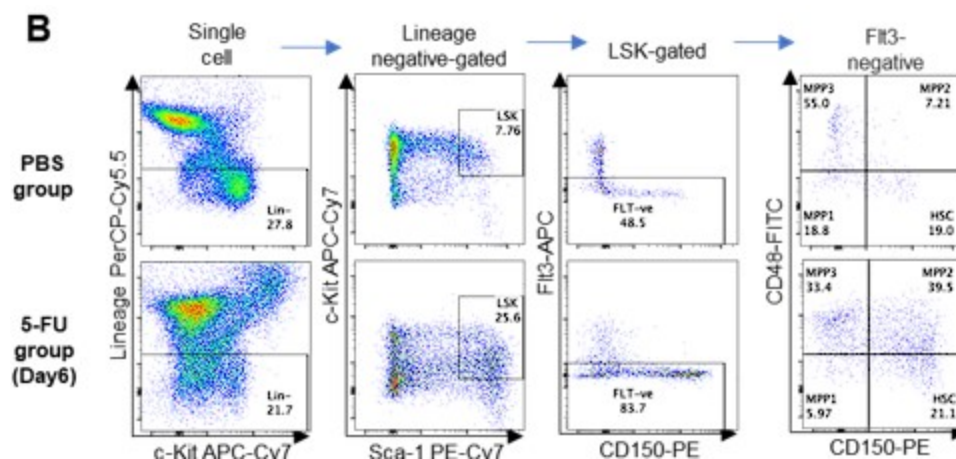
1. Confusing image label of Supplemental Figure 1H should be corrected in line 253.

We have corrected the incorrect figure caption on line 217 in the revised manuscript to "Supplemental Figure 1N" as you suggested.

1. The percentage of the indicated cell population should also be shown in Figure S1B.

As you indicated, we have included the percentages for each population in Supplemental Figure 1B.

Author response image 1.



1. Please pay attention to the small size of the marks in the graph, such as in Figure S1F and so on.

As you indicated, we have corrected the very small text contained in Figure S1F. Similar corrections have been made to Figures S1B and S5A.

1. Please pay attention to the label of line in Figure S6A-D.

Thank you very much for the advice. We have added line labels to the graph in the original Figures S6A–D.

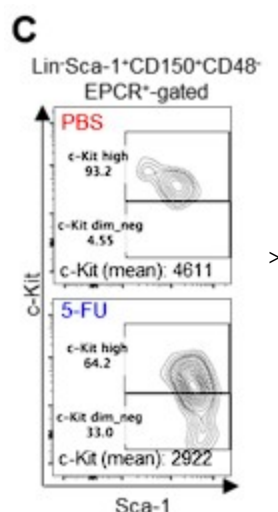
(Specific comments)

1. Based on previous reports, the authors expanded the LSK gate to include as many HSCs as possible (Supplemental Figure 1B). However, while they showed the gating strategy on Day 6 after 5-FU treatment, results from other time-points should also be displayed to ensure the strict selection of time-points.

Thank you for pointing this out. First, we did not enlarge the Sca-1 gating in this study. We apologize for any confusion caused by the incomplete description. The gating of c-Kit is based on that shown by Umemoto et al (Figure EV1A) 2, who used 250 mg/kg 5-FU, so their c-Kit reduction is more pronounced than ours.

We followed this study and compared c-Kit expression in Lin-Sca-1⁺CD150⁺CD48⁺EPCR⁺ gates to BMMNCs on day 6 after 5-FU administration (150 mg/kg). The results are shown below.

Author response image 2.

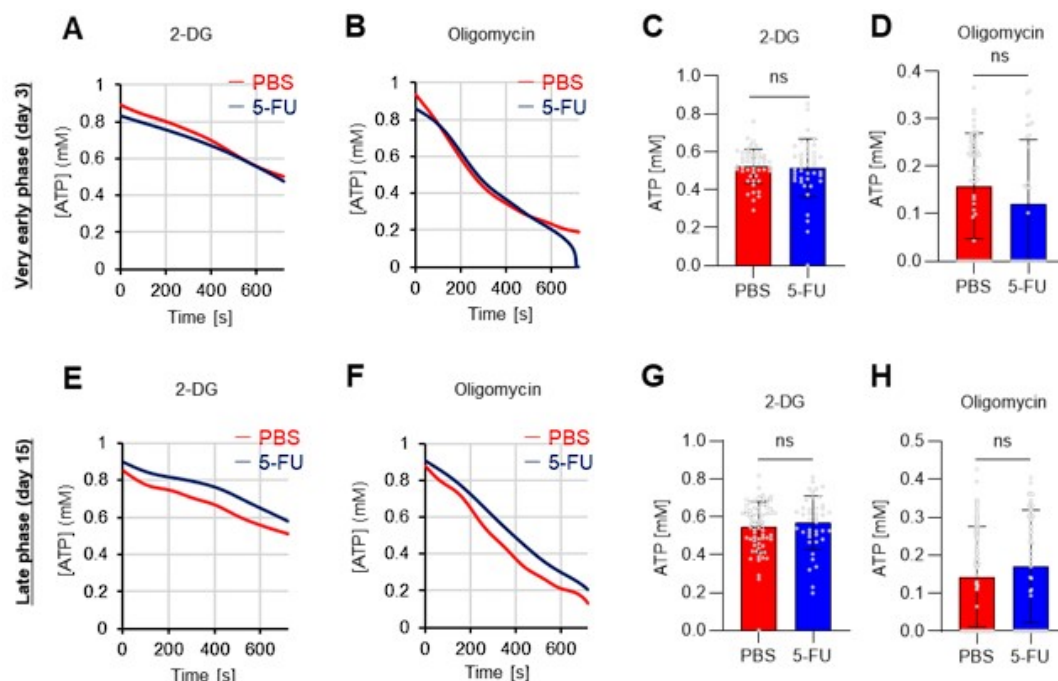


Since the MFI of c-Kit was downregulated, we used gating that extended the c-Kit gate to lower-expression regions on day 6 after 5-FU administration (revised Figure S1C). At other time points, LSK gating was the same as in the PBS-treated group, as noted in the Methods.

1. In Figure 1, the authors examined the metabolite changes on Day 6 after 5-FU treatment. However, it is important to consider whether there are any dynamic adjustments to metabolism during the early and late stages of 5-FU treatment in HSCs compared to PBS treatment, in order to coordinate cell homeostasis despite no significant changes in cell cycle progression at other time-points.

Thank you for pointing this out. Below are the results of the GO-ATeam2 analysis during the very early phase (day 3) and late phase (day 15) after 5-FU administration (revised Figures S7A–H).

Author response image 3.

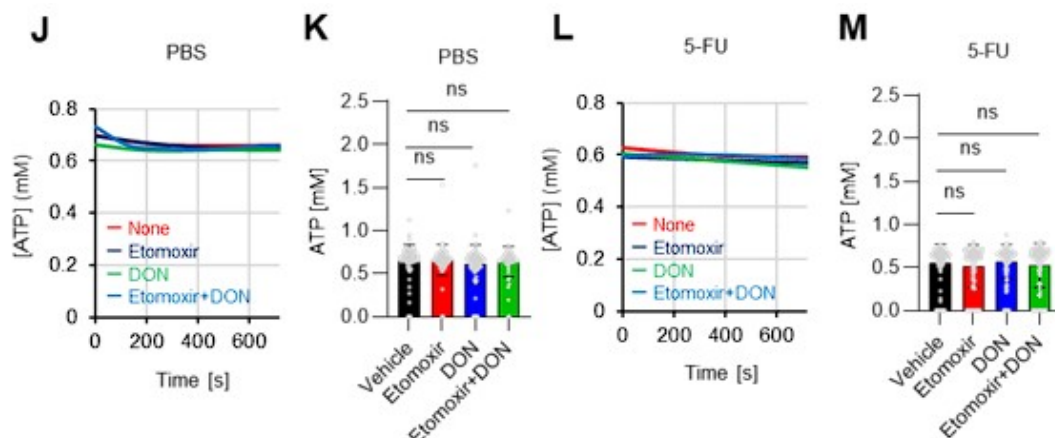


In the very early phase, such as day 3 after 5-FU administration, cell cycle progression had not started (Figure S1C) and was not preceded by metabolic changes. Meanwhile, in the late phase, such as day 15 after 5-FU administration, the cell cycle and metabolism returned to a steady state. In summary, the timing of the metabolic changes coincided with that of cell cycle progression. This point is essential for discussing the cell cycle-dependent metabolic system of HSCs and has been newly included in the Results (page 11, lines 321-323).

1. As is well known, ATP can be produced through various pathways, including glycolysis, the TCA cycle, the PPP, NAS, lipid metabolism, amino acid metabolism and so on. Therefore, it is important to investigate whether treatment with 5-FU or oligomycin affects these other metabolic pathways in HSCs.

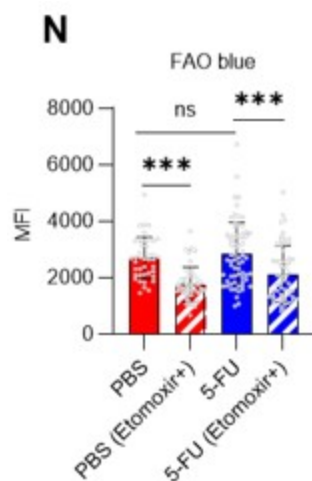
As the reviewer pointed out, ATP production by systems other than the glycolytic system of HSCs is also essential. In this revised manuscript, we examined the effects of the FAO inhibitor (Etomoxir, 100 μ M) and the glutaminolysis inhibitor 6-diazo-5-oxo-L-norleucine (DON, 2mM) alone or in combination on the ATP concentration of HSCs after PBS or 5-FU treatment. As shown below, there was no apparent decrease in ATP concentration (revised Figures S7J-M).

Author response image 4.



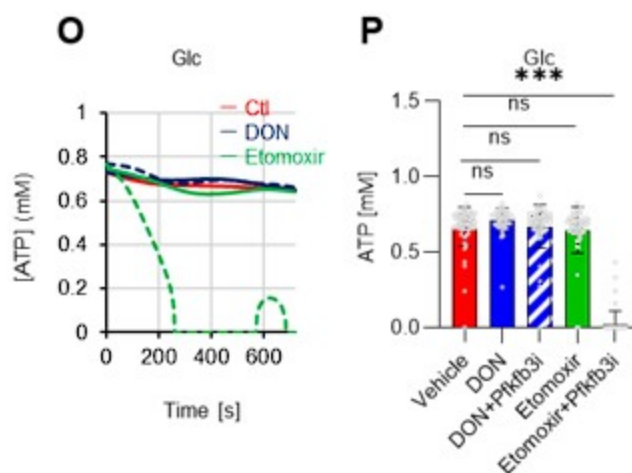
Fatty acid β -oxidation activity was also measured in 5-FU-treated HSCs using the fluorescent probe FAOBlue and was unchanged compared to PBS-treated HSCs (revised Figure S7N).

Author response image 5.



Notably, the addition of 100 μ M etomoxir plus glucose and Pfkfb3 inhibitors resulted in a rapid decrease in ATP concentration in HSCs (revised Figures S7O–P). This indicates that etomoxir partially mimics the effect of oligomycin, suggesting that at a steady state, OXPHOS is driven by FAO, but can be compensated by the acceleration of the glycolytic system by Pfkfb3. Meanwhile, the exposure of HSCs to Pfkfb3 inhibitors in addition to 2 mM DON, which is an extremely high dose considering that the K_i value of DON for glutaminase is 6 μ M, did not reduce ATP (revised Figures S7O–P). This suggests that ATP production from glutaminolysis is limited in HSCs at a steady state.

Author response image 6.



These points suggest that OXPHOS is driven by fatty acids at a steady state, but unlike the glycolytic system, FAO is not further activated by HSCs after 5-FU treatment. The results of these analyses and related descriptions are included in the revised manuscript (page 11, lines 332-344).

1. In part 2, they showed that oligomycin treatment of HSCs exhibited activation of the glycolytic system, but what about the changes in ATP concentration under oligomycin treatment? Are other metabolic systems affected by oligomycin treatment?

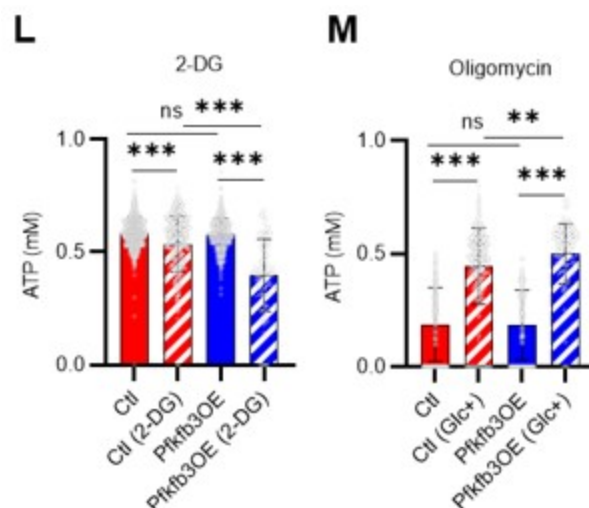
Thank you for your thoughtful comments. The relevant results we have obtained so far with the GO-ATeam2 system are as follows: First, OXPHOS inhibition in the absence of glucose significantly decreases the ATP concentration of HSCs (Figure 4C). Meanwhile, OXPHOS inhibition in the presence of glucose maintains the ATP concentration of HSCs (Figure 5B). Since it is difficult to imagine a completely glucose-free environment in vivo, it is thought that ATP concentration is maintained by the acceleration of the glycolytic system even under hypoxic or other conditions that inhibit OXPHOS.

Meanwhile, glucose tracer analysis shows that OXPHOS inhibition suppresses nucleic acid synthesis (NAS) except for the activation of the glycolytic system (Figures 2C–F). This is because phosphate groups derived from ATP are transferred to nucleotide mono-/diphosphate in NAS, but OXPHOS, the main source of ATP production, is impaired, along with the enzyme conjugated with OXPHOS in the process of NAS (dihydroorotate dehydrogenase, DHODH). We have added a new paragraph in the Discussion section (page 17, lines 511-515) to provide more insight to the reader by summarizing and discussing these points.

1. In Figure 5M, it would be helpful to include a control group that was not treated with 2-DG. Additionally, if Figure 5L is used as the control, it is unclear why the level of ATP does not show significant downregulation after 2-DG treatment. Similarly, in Figure 5O, a control group with no glucose addition should be included.

Thank you for your advice. The experiments corresponding to the control groups in Figures 5M and O were in Figures 5L and N, respectively, but we have combined them into one graph (revised Figures 5L–M). The results more clearly show that PFKFB3 overexpression enhances sensitivity to 2-DG, but also enhances glycolytic activation upon oligomycin administration.

Author response image 7.



1. In this study, their findings suggest that PFKFB3 is required for glycolysis of HSCs under stress, including transplantation. In Figure 7B, the results showed that donor-derived chimerism in PB cells decreased relative to that in the WT control group during the early phase (1 month post-transplant) but recovered thereafter. Although the transplantation cell number is equal in two groups of donor cells, it is unclear why the donor-derived cell count decreased in the 2-week post-transplantation period and recovered thereafter in the Pfkfb3 KO group. Therefore, they should provide an explanation for this. Additionally, they only detected the percentage of donor-derived cells in PB but not from BM, which makes it difficult to support the argument for Increasing the HSPC pool.

As pointed out by the reviewer, it is interesting to note that the decrease in peripheral blood chimerism in the PFKFB3 knockout is limited to immediately after transplantation and then catches up with the control group (Figure 7B). We attribute this to the fact that HSPC proliferation is delayed immediately after transplantation in PFKFB3 deficiency, but after a certain time, PB cells produced by the delayed proliferating HSPCs are supplied. In support of this, the PFKFB3 knockout HSPCs did not exhibit increased cell death after transplantation (Figure 7K), while a delayed cell cycle was observed (Figures 7G–J). A description of this point has been added to the Discussion (page 19, lines 573–579).

In addition, the knockout efficiency in bone marrow cells could not be verified because the number of cells required for KO efficiency analysis was not available. Therefore, we have added a statement on this point and have toned down our overall claim regarding the extent to which PFKFB3 is involved in the expansion of the HSPC pool (page 15, lines 474–476).

1. In Figure 7E, they collected the BM reconstructed with Pfkfb3- or Rosa-KO HSPCs two months after transplantation, and then tested their resistance to 5-FU. However, the short duration of the reconstruction period makes it difficult to draw conclusions about the effects on steady-state blood cell production.

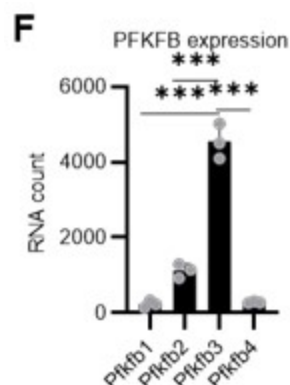
We agree that we cannot conclude from this experiment alone that PFKFB3 is completely unnecessary in steady state because, as you pointed out, the observation period of the experiment in Figure 7E is not long. We have toned down the claim by stating that PFKFB3 is

only less necessary in steady-state HSCs compared to proliferative HSCs (page 15, lines 460-461).

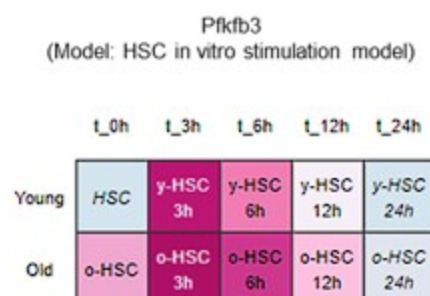
1. PFK is allosterically activated by PFKFB, and other members of the PFKFB family could also participate in the glycolytic program. Therefore, they should investigate their function in contributing to glycolytic plasticity in HSCs during proliferation. Additionally, they should also analyze the protein expression and modification levels of other members. Although PFKFB3 is the most favorable for PFK activation, the role of other members should also be explored in HSC cell cycling to provide sufficient reasoning for choosing PFKFB3.

To further justify why we chose PFKFB3 among the PFKFB family members, we reviewed our data and the publicly available Gene Expression Commons (GEXC) 3. PFKFB3 is the most highly expressed member of the PFKFB family in HSCs (revised Figure 4F), and its expression increases with proliferation (Author response image 9). In addition to this, we have also cited the literature 4 indicating that AZ PFKFB3 26 is a Pfkfb3-specific inhibitor that we used in this paper, and added a note to this point (that it is specific) (page 11, lines 327-329). Through these revisions, we sought to strengthen the rationale for Pfkfb3 as the primary target of the analysis.

Author response image 8.



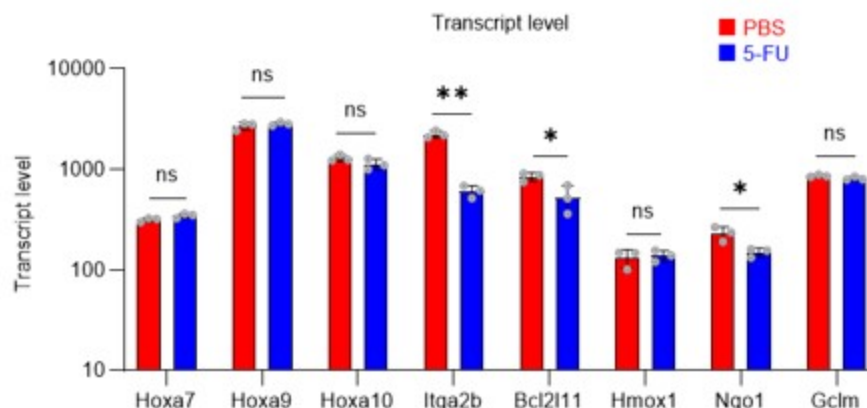
Author response image 9.



1. In this study, the authors identified PRMT1 as the upstream regulator of PFKFB3 that is involved in the glycolysis activation of HSCs. However, PRMT1 is also known to participate in various transcriptional activations. Thus, it is important to determine whether PRMT1 affects glycolysis through transcriptional regulation or through its direct regulation of PFKFB3? Additionally, the authors should investigate whether PRMT1i inhibits ATP production in normal HSCs. Moreover, could we combine Figure 6I and 6J for analysis. Finally, the authors could conduct additional rescue experiments to demonstrate that the effect of PRMT1 inhibitors on ATP production can be rescued by overexpression of PFKFB3.

Although PRMT1 inhibition reduced m-PFKFB3 levels in HSCs, 5-FU treatment also reduced or did not alter Pfkfb3 transcript levels (Figures 6B, G) and the expression of genes such as Hoxa7/9/10, Itga2b, and Nqo1, which are representative transcriptional targets of PRMT1, in proliferating HSCs after 5-FU treatment (revised Figure S9).

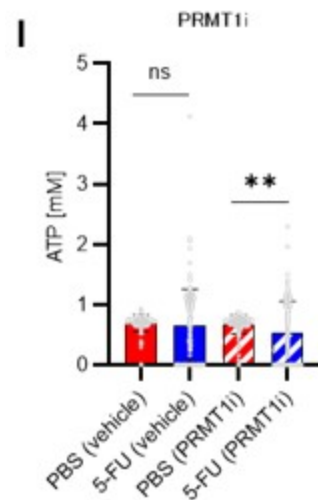
Author response image 10.



These results suggest that PRMT1 promotes PFKFB3 methylation, which increases independently of transcription in HSCs after 5-FU treatment.

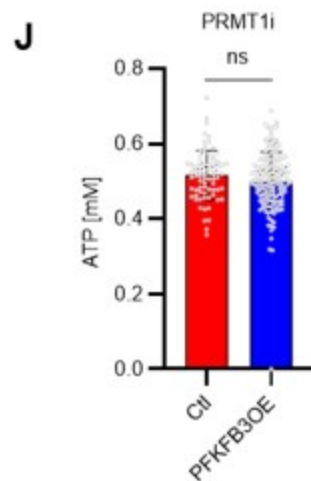
A summary analysis of the original Figures 6I and 6J is shown below (revised Figure 6I).

Author response image 11.



Finally, we tested whether the inhibition of the glycolytic system and the decrease in ATP concentration due to PRMT1 inhibition could be rescued by the retroviral overexpression of PFKFB3. We found that PFKFB3 overexpression did not decrease the ATP concentration in HSCs due to PRMT1 inhibition (revised Figure 6J). Therefore, PFKFB3 overexpression mitigated the decrease in ATP concentration caused by PRMT1 inhibition. These data and related statements have been added to the revised manuscript (page 14, lines 427-428).

Author response image 12.



Reviewer #2:

In the manuscript Watanuki et al. want to define the metabolic profile of HSCs in stress/proliferative (myelosuppression with 5-FU), and mitochondrial inhibition and homeostatic conditions. Their conclusions are that during proliferation HSCs rely more on glycolysis (as other cell types) while HSCs in homeostatic conditions are mostly dependent on mitochondrial metabolism. Mitochondrial inhibition is used to demonstrate that blocking mitochondrial metabolism results in similar features of proliferative conditions.

The authors used state-of-the-art technologies that allow metabolic readout in a limited number of cells like rare HSCs. These applications could be of help in the field since one of the major issues in studying HSCs metabolism is the limited sensitivity of the "standard" assays, which make them not suitable for HSC studies.

However, the observations do not fully support the claims. There are no direct evidence/experiments tackling cell cycle state and metabolism in HSCs. Often the observations for their claims are indirect, while key points on cell cycle state-metabolism, OCR analysis should be addressed directly.

We sincerely appreciate the reviewer's constructive comments. Thank you for highlighting the importance of the highly sensitive metabolic assay developed in this study and the findings based on it. Meanwhile, the reviewer's comments have made us aware of areas where we can further improve this manuscript. In particular, in the revised manuscript, we have performed further studies to demonstrate the link between the cell cycle and metabolic state. Specifically, we further subdivided HSCs by the uptake of in vivo-administered 2-NBDG and performed cell cycle analysis. Next, HSCs after PBS or 5-FU treatment were analyzed by a Mito Stress test using the Seahorse flux analyzer, including ECAR and OCR, and a more direct relationship between the cell cycle state and the metabolic system was found. We believe that the reviewer's valuable suggestions have helped us clarify more directly the importance of the metabolic state of HSCs in response to cell cycle and stress that we wanted to show and emphasize the usefulness of the GO-ATeam2 system. Our response to "Recommendations For The Authors" is listed first, followed by our responses to all comments in "Public Review" as follows:

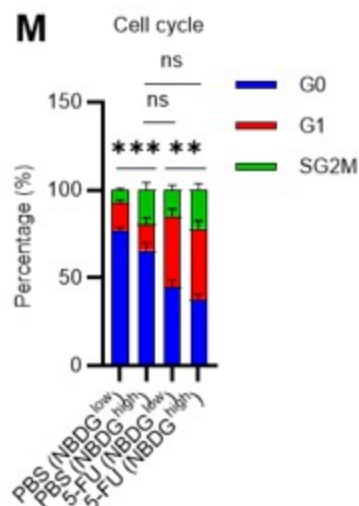
(Recommendations For The Authors):

In general, I believe it would be important:

- 1. to directly associate cell cycle state with metabolic state. For example, by sorting HSC (+/- 5FU) based on their cell cycle state (exploiting the mouse model presented in the manuscript or by defining G0/G1/G2-S-M via Pyronin/Hoechst staining which allow to sort live cells) and follow the fate of radiolabeled glucose.*

Thank you for raising these crucial points. Unfortunately, it was difficult to perform the glucose tracer analysis by preparing HSCs with different cell cycle states as you suggested due to the amount of work involved. In particular, in the 5-FU group, more than 60 mice per group were originally required for an experiment, and further cell cycle-based purification would require many times that number of mice, which we felt was unrealistic under current technical standards. As an alternative, we administered 2-NBDG to mice and fractionated HSCs at the 2-NBDG fluorescence level for cell cycle analysis. The results are shown below (revised Figure S1M). Notably, even in the PBS-treated group, HSCs with high 2-NBDG uptake were more proliferative than those with low 2-NBDG uptake and are comparable to HSCs after 5-FU treatment, although the overall population of HSCs exiting the G0 phase and entering the G1 phase increased after 5-FU treatment. In both PBS/5-FU-treated groups, these large differences in cell cycle glucose utilization suggest a direct link between HSC proliferation and glycolysis activation. If a more sensitive type of glucose tracer analysis becomes available in the future, it may be possible to directly address the reviewer's comments. We see this as a topic for the future. The descriptions of the above findings and perspectives have been added to the Results and Discussion section (page 7, lines 208-214, page 20, lines 607-610).

Author response image 13.

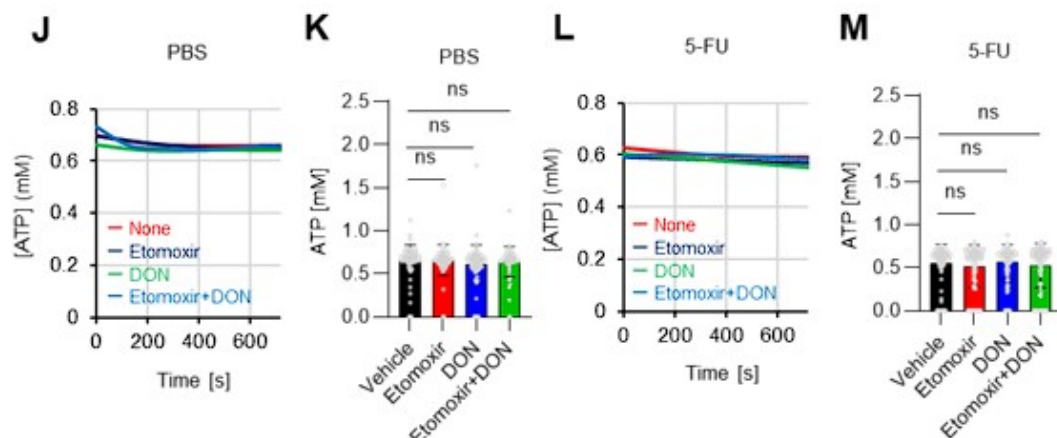


1. Use other radio labeled substrates (fatty acid, glutamate)

Thank you very much for your suggestion. While this is an essential point for future studies, we believe it is not the primary focus of the paper. We are planning another research project on tracer analysis using labeled fatty acids and glutamates, which we will report on in the near future. We have clearly stated in the Abstract and Introduction of the revised manuscript, that the focus of this study is on changes in glucose metabolism when HSCs are stressed (page 3, line 75 and 87, page 5, lines 135).

Instead, we added the following analyses of metabolic changes in fatty acids and glutamate using the GO-Ateam2 system. HSCs derived from GO-Ateam2 mice treated with PBS or 5-FU were used to measure changes in ATP concentrations after exposure to the fatty acid beta-oxidation (FAO) inhibitor etomoxir and the glutaminolysis inhibitor 6-diazo-5-oxo-L-norleucine (DON). Etomoxir was used at 100 μ M, a concentration that inhibits FAO without inhibiting mitochondrial electron transfer complex I, as previously reported 5. DON was used at 2 mM, a concentration that sufficiently inhibits the enzyme as the K_i for glutaminase is 6 μ M. In this experiment, etomoxir alone, DON alone, or etomoxir and DON in combination did not decrease the ATP concentration of HSCs in the PBS and 5-FU groups (revised Figures S7J–M), suggesting that FAO and glutaminolysis were not essential for ATP production in HSCs in the short term. Thus, according to the analysis using the GO-Ateam2 system, HSCs exposed to acute stresses change the efficiency of glucose utilization (accelerated glycolytic ATP production) rather than other energy sources. Since there are reports that FAO and glutaminolysis are required for HSC maintenance in the long term 5,6, compensatory pathways may be able to maintain ATP levels in the short term. A description of these points has been added to the Discussion (page 11, lines 332-344).

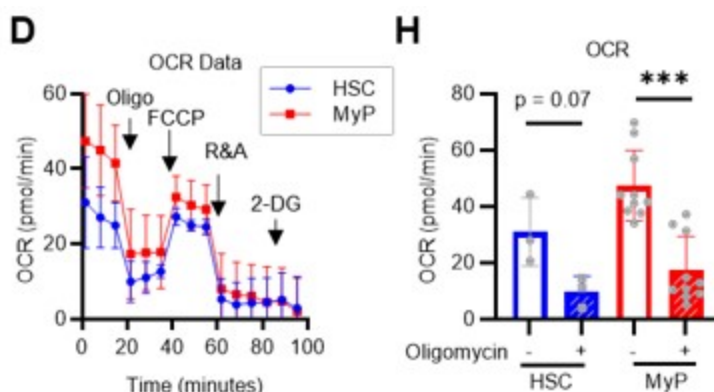
Author response image 14.



1. Include OCR analyses.

In addition to the ECAR data of the Mito Stress test (original Figures 2G–H), OCR data were added to the revised manuscript (revised Figures 2H, S3D). Compared to c-Kit⁺ myeloid progenitors (LKS- cells), HSC showed a similar increase in ECAR, while the decrease in OCR was relatively limited. A possible explanation for this is that glycolytic and mitochondrial metabolism are coupled in c-Kit⁺ myeloid progenitors, whereas they are decoupled in HSCs. This is also suggested by the glucose plus oligomycin experiment in Figures 5B, C, and S6A–D (orange lines). In summary, in HSCs, glycolytic and mitochondrial ATP production are decoupled and can maintain ATP levels by glycolytic ATP production alone, whereas in progenitors including GMPs, the two ATP production systems are constantly coupled, and glycolysis alone cannot maintain ATP concentration. We have added descriptions of these points in the Results and Discussion section (page 8, lines 240–243, page 18, lines 558–561).

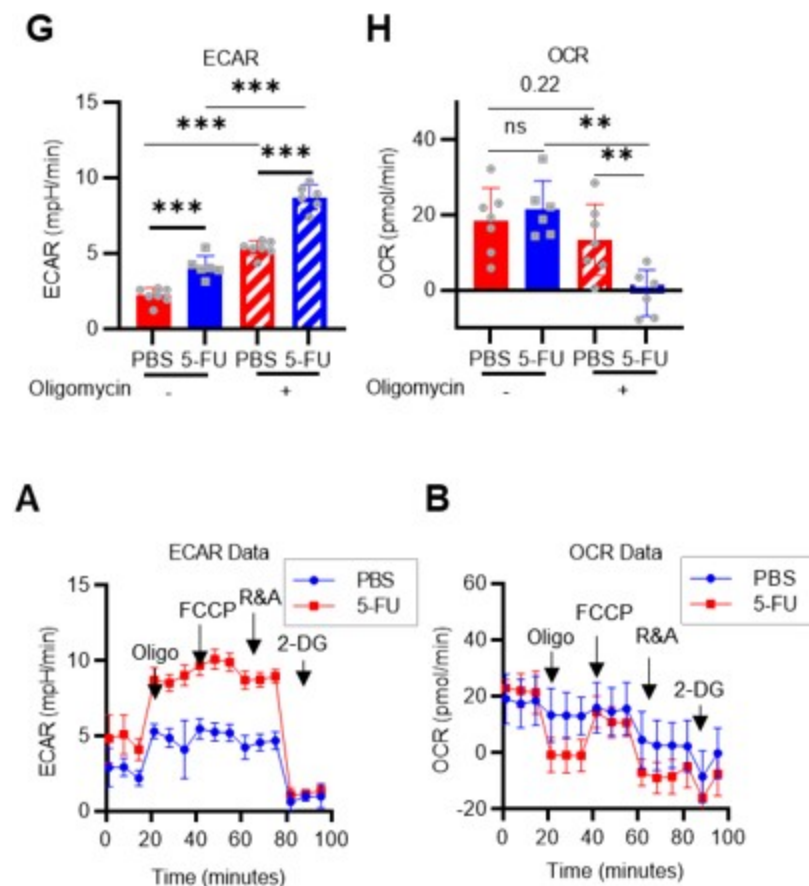
Author response image 15.



Next, a Mito Stress test was performed using HSCs derived from PBS- or 5-FU-treated mice in the presence or absence of oligomycin (revised Figures 1G–H, S3A–B). Without oligomycin treatment, ECAR in 5-FU-treated HSCs was higher than in PBS-treated HSCs, and OCR was unchanged. Oligomycin treatment increased ECAR in both PBS- and 5-FU-treated HSCs, whereas OCR was unchanged in PBS-treated HSCs, but significantly decreased in 5-FU-treated HSCs. Changes in ECAR in response to oligomycin differed between HSC proliferation or

differentiation: ECAR increased in 5-FU-treated HSCs but not in LKS- progenitors (original Figures 2G–H). This suggests a metabolic feature of HSCs in which the coupling of OXPHOS with glycolysis seen in LKS- cells is not essential in HSCs even after cell cycle entry. The results and discussion of this experiment have been added to page 7, lines 194-201 and page 18, lines 558-561).

Author response image 16.



1. Correlate proliferation-mitochondrial inhibition-metabolic state

We agree that it is important to clarify this point. First, OXPHOS inhibition and proliferation similarly accelerate glycolytic ATP production with PFKFB3 (Figures 4G, I, and 5F–I). Meanwhile, oligomycin treatment rapidly decreases ATP in HSCs with or without 5-FU administration (Figure 4C). These results suggest that OXPHOS is a major source of ATP production both at a steady state and during proliferation, even though the analysis medium is pre-saturated with hypoxia similar to that in vivo. This has been added to the Discussion section (page 17, lines 520-523).

1. Tune down the claim on HSCs in homeostatic conditions since from the data it seems that HSCs rely more on anaerobic glycolysis.

Thanks for the advice. The original Figures S2C, D, F, and G show that HSC is dependent on the anaerobic glycolytic system even at a steady state, so we have toned down our claims (page 7, lines 192-194).

1. For proliferative HSCs mitochondria are key. When you block mitochondria with oligomycin there's the biggest drop in ATP.

In the revised manuscript, we have tried to highlight the key findings that you have pointed out. First, we mentioned in the Discussion (page 17, lines 523-525) that previous studies suggested the importance of mitochondria in proliferating HSCs. Meanwhile, the GO-ATeam2 and glucose tracer analyses in this study newly revealed that the glycolytic system activated by PFKFB3 is activated during the proliferative phase, as shown in Figure 4C. We also confirmed that mitochondrial ATP production is vital in proliferating HSCs, and we hope to clarify the balance between ATP-producing pathways and nutrient sources in future studies.

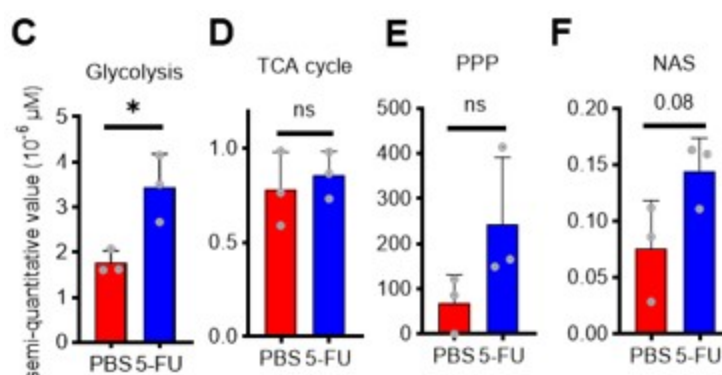
1. To better clarify this point authors, authors should do experiments in hypoxic conditions and compare it to oligomycin treatment and showing that mito-inhibition acts differently on HSCs (considering that all these drugs are toxic for mitochondria and induce rapidly stress responses ex: mitophagy).

We apologize for any confusion caused by not clearly describing the experimental conditions. As pointed out by the reviewer, we also recognize the importance of experiments in a hypoxic environment. All GO-ATeam2 analyses were performed in a medium saturated sufficiently under hypoxic conditions and analyzed within minutes, so we believe that the medium did not become oxygenated (page S5-S6, lines 160-163 in the Methods). Despite being conducted under such hypoxic conditions, the substantial decrease in ATP after oligomycin treatment is intriguing (original Figures 4C, 5B, 5C). The p50 value of mitochondria (the partial pressure of oxygen at which respiration is half maximal) is 0.1 kPa, which is less than 0.1% of the oxygen concentration at atmospheric pressure 7. Thus, biochemically, it is consistent that OXPHOS can maintain sufficient activity even in a hypoxic environment like the bone marrow. We are currently embarking on a study to determine ATP concentration in physiological hypoxic conditions using in vivo imaging within the bone marrow, which we hope to report in a separate project. We have discussed these points, technical limitations, and perspectives in the Discussion section (page 20, lines 610-612).

• In Figure 1 C, D, E and F, the comparison should be done as unpaired t test and the control group should not be 1 as the cells comes from different individuals.

Thank you very much for pointing this out. We have reanalyzed and revised the figures (revised Figures 1C-F)

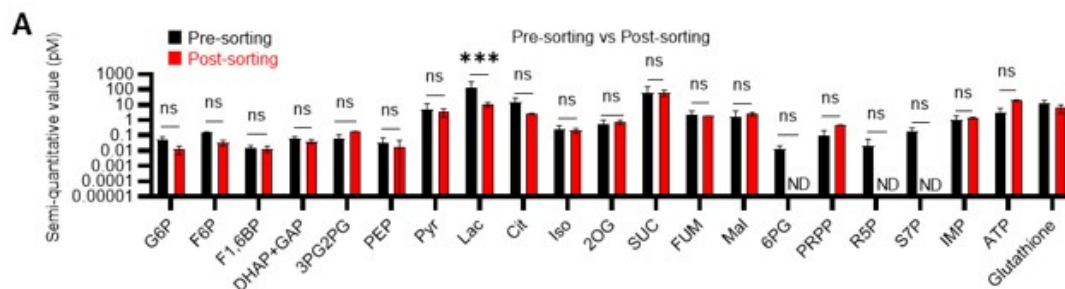
Author response image 17.



• In Figure S2A, the post-sorting bar of 6PG, R5P and S7P are missing.

Metabolites below the detection threshold (post-sorting samples of 6PG, R5P, and S7P) are now indicated as N.D. (not detected) (revised Figure S2A).

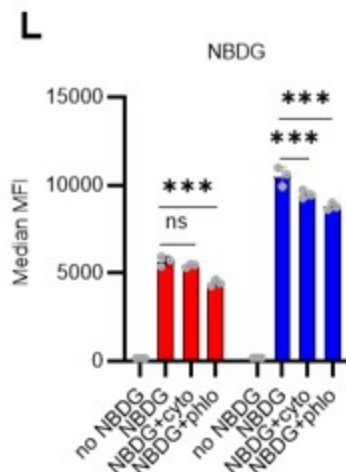
Author response image 18.



• In the 2NBDG experiments, authors should add the appropriate controls, since it has been shown that 2NBDG cellular uptake do not correctly reflect glucose uptake (Sinclair LV, Immunometabolism 2020) (a cell type dependent variations) thus inhibitors of glucose transporters should be added as controls (cytochalasin B; 4,6-O-ethylidene- α -D-glucose) it would be quite challenging to test it in vivo but it would be sufficient to show that in vitro in the different HSPCs analyzed.

We appreciate the essential technical point raised by the reviewer. In the revised manuscript, we performed a 2-NBDG assay with cytochalasin B and phloretin as negative controls. After PBS treatment, 2-NBDG uptake was higher in 5-FU-treated HSCs compared to untreated HSCs. This increase was inhibited by both cytochalasin B and phloretin. In PBS-treated HSCs, cytochalasin B did not downregulate 2-NBDG uptake, whereas phloretin did. Although cytochalasin B inhibits glucose transporters (GLUTs), it is also an inhibitor of actin polymerization. Therefore, its inhibitory effect on GLUTs may be weaker than that of phloretin. We have revised the figure (revised Figure S1L) and added the corresponding description (page 7, lines 207-208).

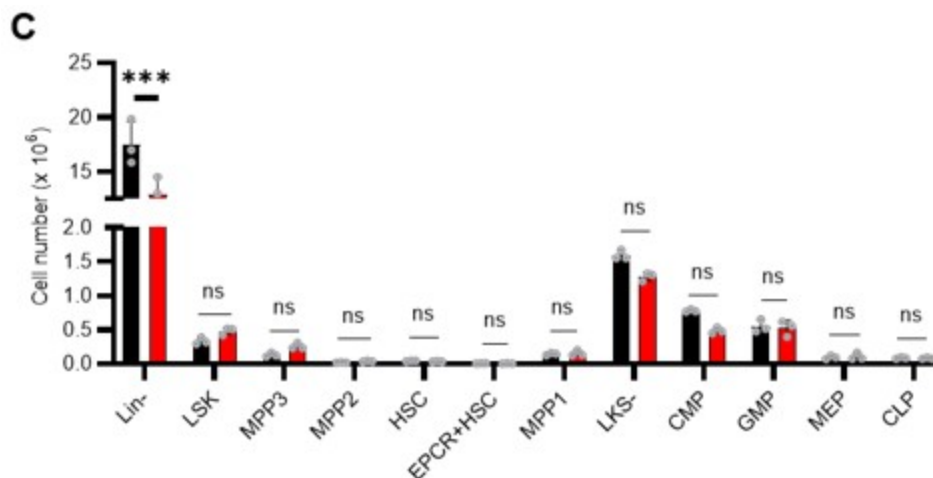
Author response image 19.



• S5C: authors should show the cell number for each population. If there's a decreased in % in Lin⁻ that will be reflected in all HSPCs. Comparing the proportion of the cells doesn't show the real impact on HSPCs.

Thank you for your insightful point. In the revision, we compared the numbers, not percentages, of HSPCs and found no difference in the number of cells in the major HSPC fractions in Lin⁻. The figure has been revised (revised Figure S6C) and the corresponding description has been added (page 10, lines 296-299).

Author response image 20.



Minor:

1. In S1 F-G is not indicated in which day post 5FU injection is done the analysis. I assume on day 6 but it should be indicated in the figure legend and/or text.

Thank you for pointing this out. As you assumed, the analysis was performed on day 6. The description has been added to the legend of the revised Figure S1G.

1. S1K is not described in the text. What are proliferative and quiescence-maintaining conditions? The analyses are done by flow using LKS SLAM markers after culture? How long was the culture?

Thank you for your comments. First, the figure citation on line 250 was incorrect and has been corrected to Figure S1N. Regarding the proliferative and quiescence-maintaining conditions, we have previously reported on these 8. In brief, these are culture conditions that maintain HSC activity at a high level while allowing for the proliferation or maintenance of HSCs in quiescence, achieved by culturing under fatty acid-rich, hypoxic conditions with either high or low cytokine concentrations. Analysis was performed after one week of culture, with the HSC number determined by flow cytometry based on the LSK-SLAM marker. While these are mentioned in the Methods section, we have added a description in the main text to highlight these points for the reader (page 7, lines 214-217).

1. In Figure 5G, why does the blue line (PFKFB3 inhibitor) go up in the end of the real-time monitoring? Does it mean that other compensatory pathway is turned on?

As you have pointed out, we cannot rule out the possibility that other unknown compensatory ATP production pathways were activated. We have added a note in the Discussion section to address this (page 18, lines 555-556).

1. In Figure S6H&J, the reduction is marginal. Does it mean that PKM2 is not important for ATP production in HSCs?

The activity of the inhibitor is essential in the GO-ATeam2 analysis. The commercially available PKM2 inhibitors have a higher IC₅₀ value (IC₅₀ = 2.95 μ M in this case). Nevertheless, the effect of reducing the ATP concentration was observed in progenitor cells, but not in HSCs. The report by Wang et al. 9 on the analysis using a PKM2-deficient model suggests a stronger effect on progenitor cells than on HSCs. Our results are similar to those of the previous report.

(Specific comments)

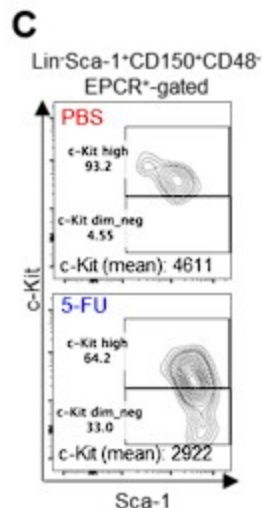
Specifically, there are several major points that rise concerns about the claims:

1. *The gating strategy to select HSCs with enlarged Sca1 gating is not convincing. I understand the rationale to have a sufficient number of cells to analyze, however this gating strategy should be applied also in the control group. From the FACS plot seems that there are more HSCs upon 5FU treatment (Figure S1b). How that is possible? Is it because of the 20% more of cycling cells at day 6? To prove that this gating strategy still represents a pure HSC population, authors should compare the blood reconstitution capability of this population with a "standard" gated population. If the starting population is highly heterogeneous then the metabolic readout could simply reflect cell heterogeneity.*

Thank you for pointing this out. First, we did not enlarge the Sca-1 gating in this study. We apologize for any confusion caused by the incomplete description. The gating of c-Kit is based on that shown by Umemoto et al (Figure EV1A) 2, who used 250 mg/kg 5-FU, so their c-Kit reduction is more pronounced than ours.

We followed this study and compared c-Kit expression in the Lin-Sca-1+CD150+CD48-EPCR+ gates to BMMNCs on day 6 after 5-FU administration (150 mg/kg). The results are shown below.

Author response image 21.



Since the MFI of c-Kit was downregulated, we used gating that extended the c-Kit gate to lower expression regions on day 6 after 5-FU administration (revised Figure S1C).

At other time points, LSK gating was the same as in the PBS-treated group, as noted in the Methods.

The reason why the number of HSCs appears to be higher in the 5-FU group is because most of the differentiated blood cells were lost due to 5-FU administration and the same number of cells as in the PBS group were analyzed by FACS, resulting in a relatively higher number of HSCs. The legend of Figure S1 shows that the number of HSCs in both the PBS and 5-FU groups appeared to increase because the same number of BMMNCs was obtained at the time of analysis (page S22, lines 596-598).

Regarding cellular heterogeneity, from a metabolic point of view, the heterogeneity in HSCs is rather reduced by 5-FU administration. As shown in Figure S3A–C, this is simulated under stress conditions, such as after 5-FU administration or during OXPHOS inhibition, where the flux variability in each enzymatic reaction is significantly reduced. GO-ATeam2 analysis after 5-FU treatment showed no increase in cell population variability. After 2-DG treatment, ATP concentrations in HSCs were widely distributed from 0 mM to 0.8 mM in the PBS group, while more than 80% of those in the 5-FU group were less than 0.4 mM (Figures 4B, D). HSCs may have a certain metabolic diversity at a steady state, but under stress conditions, they may switch to a more specialized metabolism with less cellular heterogeneity in order to adapt.

1. S2 does not show major differences before and after sorting. However, a key metabolite like Lactate is decreased, which is also one of the most present. Wouldn't that mean that HSCs once they move out from the hypoxic niche, they decrease lactate production? Do they decrease anaerobic glycolysis? How can quiescent HSC mostly rely on OXPHOS being located in hypoxic niche?

1. Since HSCs in the niche are located in hypoxic regions of the bone marrow, would that not mimic OxPhos inhibition (oligomycin)? Would that not mean that HSCs in the niche are more glycolytic (anaerobic glycolysis)?

1. In Figure 5B, the orange line (Glucose+OXPHOS inhibition) remains stable, which means HSCs prefer to use glycolysis when OXPHOS is inhibited. Which metabolic pathway would HSCs use under hypoxic conditions? As HSCs reside in hypoxic niche, does it mean that these steady-state HSCs prefer to use glycolysis for ATP production? As mentioned before, mitochondrial inhibition can be comparable at the in vivo condition of the niche, where low pO₂ will "inhibit" mitochondria metabolism.

Thank you for the first half of comment 2 on the technical features of our approach. First, as you have pointed out, there is minimal variation and stable detection of many metabolites before and after sorting (Figure S2A), suggesting that isolation from the hypoxic niche and sorting stress do not significantly alter metabolite detection performance. This is consistent with a previous report by Jun et al. 10. Meanwhile, lactate levels decreased by sorting. Therefore, if the activity of anaerobic glycolysis was suppressed in stressed HSCs, it may be difficult to detect these metabolic changes with our tracer analysis. However, in this study, several glycolytic metabolites, including an increase in lactate, were detected in HSCs from 5-FU-treated mice compared with HSCs from PBS-treated mice that were similarly sorted and prepared, suggesting an increase in glycolytic activity. In other words, we may have been fortunate to detect the stress-induced activation of the glycolytic system beyond the characteristic of our analysis system that lactate levels tend to appear lower than they are. Given that damage to the bone marrow hematopoiesis tends to alleviate the low-oxygen status of the niche 11, we postulate that this upregulated aerobic glycolysis arises intrinsically in HSCs rather than from external conditions.

The second half of comment 2, and comments 7 and 10, are essential and overlapping comments and will be answered together. Although genetic analyses have shown that HSCs produce ATP by anaerobic glycolysis in low-oxygen environments 9,12, our GO-ATeam2 analysis in this study confirmed that HSCs also generate ATP via mitochondria. This is also supported by Ansó's prior findings where the knockout of the Rieske iron-sulfur protein (RISP), a constituent of the mitochondrial electron transport chain, impairs adult HSC quiescence and bone marrow repopulation 13. Bone marrow is a physiologically hypoxic environment (9.9–32.0 mmHg 11). However, the p₅₀ value of mitochondria (the partial pressure of oxygen at which respiration is half maximal) is below 0.1% oxygen concentration at atmospheric pressure (less than 1 mmHg) 7. This suggests that OXPHOS can retain sufficient activity even under physiologically hypoxic conditions. We are currently initiating efforts to discern ATP concentrations in vivo within the bone marrow under physiological hypoxia. This will be reported in a separate project in the future. Admittedly, when we began this research, we did not anticipate the significant mitochondrial reliance of HSCs. As we previously reported, the metabolic uncoupling of glycolysis and mitochondria 12 may enable HSCs to activate only glycolysis, and not mitochondria, under stress conditions such as post-5-FU administration, suggesting a unique metabolic trait of HSCs. We have included these technical limitations and perspectives in the Discussion section (page 17, lines 520-523).

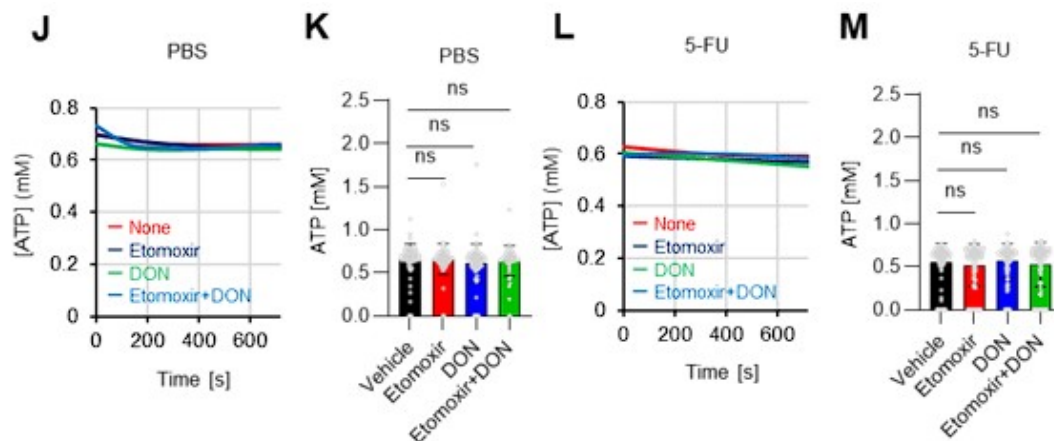
1. The authors performed challenging experiments to track radiolabeled glucose, which are quite remarkable. However, the data do not fully support the conclusions. Mitochondrial metabolism in HSCs can be supported by fatty acid and glutamate, thus authors should track the fate of other energy sources to fully discriminate the glycolysis vs mitochondrial metabolism dependency. From the data on S2 and Fig1 1C-F, the authors can conclude that upon 5FU treatment HSCs increase glycolytic rate.

1. FIG.2B-C: Increase of Glycolysis upon oligomycin treatment is common in many different cell types. As explained before, other radiolabeled substrates should be used to understand the real effect on mitochondria metabolism.

Thank you for your suggestion. While this is essential for future studies, we believe it is not the primary focus of the paper. We are planning another research project on tracer analysis using labeled fatty acids and glutamates, which we will report on in the near future. We have clearly stated in the Abstract and Introduction of the revised manuscript that the focus of this study is on changes in glucose metabolism when HSCs are stressed (page 3, line 75 and 87, page 5, lines 135).

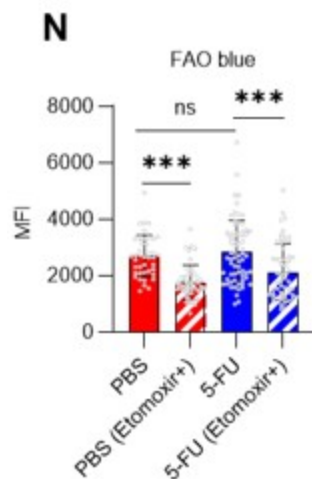
Instead, we have added the following analyses of metabolic changes in fatty acids and glutamate using the GO-ATeam2 system: HSCs derived from GO-ATeam2 mice treated with PBS or 5-FU were used to measure changes in ATP concentrations after exposure to the fatty acid beta-oxidation (FAO) inhibitor etomoxir and the glutaminolysis inhibitor 6-diazo-5-oxo-L-norleucine (DON). Etomoxir was used at 100 μ M, a concentration that inhibits FAO without inhibiting mitochondrial electron transfer complex I, as previously reported 5. DON was used at 2 mM, a concentration that sufficiently inhibits the enzyme as the K_i for glutaminase is 6 μ M. In this experiment, etomoxir alone, DON alone, or etomoxir and DON in combination did not decrease the ATP concentration of HSCs in the PBS and 5-FU groups (revised Figures S7J–M), suggesting that FAO and glutaminolysis were not essential for ATP production in HSCs in the short term. Thus, according to the analysis using the GO-Ateam2 system, HSCs exposed to acute stresses change the efficiency of glucose utilization (accelerated glycolytic ATP production) rather than other energy sources. Since there are reports that FAO and glutaminolysis are required for HSC maintenance in the long term 5,6, compensatory pathways may be able to maintain ATP levels in the short term. A description of these points has been added to the Discussion (page 17, lines 525-527).

Author response image 22.



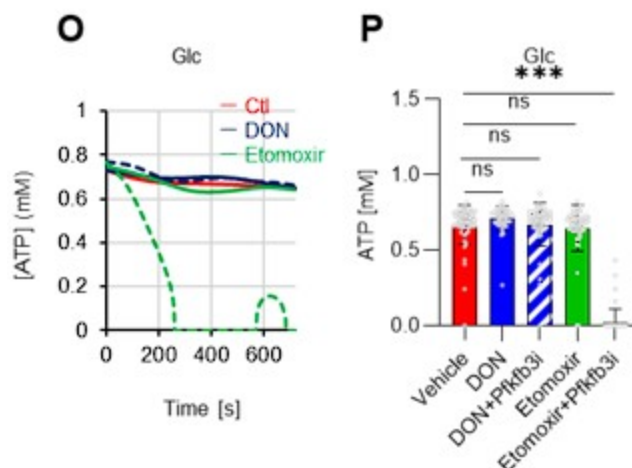
Fatty acid β -oxidation activity was also measured in 5-FU-treated HSCs using the fluorescent probe FAOBlue and was unchanged compared to PBS-treated HSCs (revised Figure S7N).

Author response image 23.



Notably, the addition of 100 μ M etomoxir plus glucose and Pfkfb3 inhibitors resulted in a rapid decrease in ATP concentration in HSCs (revised Figures S7O–P). This indicates that etomoxir partially mimics the effect of oligomycin, suggesting that at a steady state, OXPHOS is driven by FAO, but can be compensated by the acceleration of the glycolytic system by Pfkfb3. Meanwhile, the exposure of HSCs to Pfkfb3 inhibitors in addition to 2 mM DON did not reduce ATP (revised Figures S7O–P). This suggests that ATP production from glutaminolysis is limited in HSCs at a steady state.

Author response image 24.

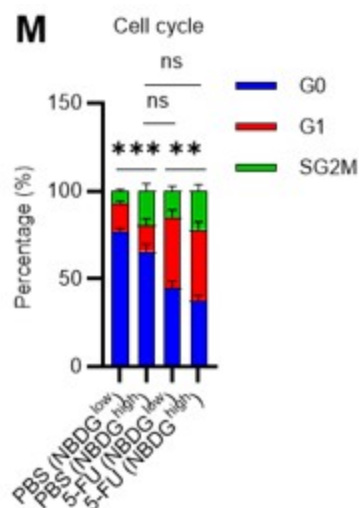


These points suggest that OXPHOS is driven by fatty acids at a steady state, but unlike the glycolytic system, FAO is not further activated by HSCs after 5-FU treatment. The results of these analyses and related descriptions are included in the revised manuscript (page 11, lines 332–344).

1. In Figure S1, 5-FU leads to the induction of cycling HSCs and in figure 1, 5-FU results in higher activation of glycolysis. Would it be possible to correlate these two phenotypes together? For example, by sorting NBDG+ cells and checking the cell cycle status of these cells?

We appreciate the reviewer's insightful comments. We administered 2-NBDG to mice and fractionated HSCs at the 2-NBDG fluorescence level for cell cycle analysis. The results are shown below (revised Figure S1M). Notably, even in the PBS-treated group, HSCs with high 2-NBDG uptake were more proliferative than HSCs with low 2-NBDG uptake and were comparable to HSCs after 5-FU treatment, although the overall population of HSCs that exited the G0 phase and entered the G1 phase increased after 5-FU treatment. In both PBS/5-FU-treated groups, these profound differences in cell cycle glucose utilization suggest a direct link between HSC proliferation and glycolysis activation. Descriptions of the above findings and perspectives have been added to the Results and Discussion section (page 7, lines 208-214, page 20, lines 607-610).

Author response image 25.



1. Why are only ECAR measurements (and not OCR measurements) shown? In Fig.2G, why are HSCs compared with cKit+ myeloid progenitors, and not with MPP1? The ECAR increased observed in HSC upon oligomycin treatment is shared with many other types of cells. However, cKit+ cells have a weird behavior. Upon oligo treatment cKit+ cells decrease ECAR, which is quite unusual. The data of both HSCs and cKit+ cells could be clarified by adding OCR curves. Moreover, it is recommended to run glycolysis stress test profile to assess the dependency to glycolysis (Glucose, Oligomycin, 2DG).

In addition to the ECAR data of the Mito Stress test (original Figures 2G–H), OCR data were added in the revised manuscript (revised Figures 2H, S3D). Compared to c-Kit+ myeloid progenitors (LKS- cells), HSC exhibited a similar increase in ECAR, while the decrease in OCR was relatively limited. This may be because glycolytic and mitochondrial metabolism are coupled in c-Kit+ myeloid progenitors, whereas they are decoupled in HSCs. This is also suggested by the glucose plus oligomycin experiment in Figures 5B, C, and S6A–D (orange lines). In summary, in HSCs, glycolytic and mitochondrial ATP production are decoupled and can maintain ATP levels by glycolytic ATP production alone, whereas in progenitors including GMPs, the two ATP production systems are constantly coupled, and glycolysis alone cannot

maintain the ATP concentration. While we could not conduct a glycolysis stress test, we believe that Pfkfb3-dependent glycolytic activation, which is evident in the oligomycin+glucose+Pfkfb3i experiment, is only apparent in HSCs when subjected to glucose+oligomycin treatment (original Figures 5F–I). We have added descriptions of these points in the Results and Discussion section (page 8, lines 240–243, page 18, lines 558–561).

Author response image 26.

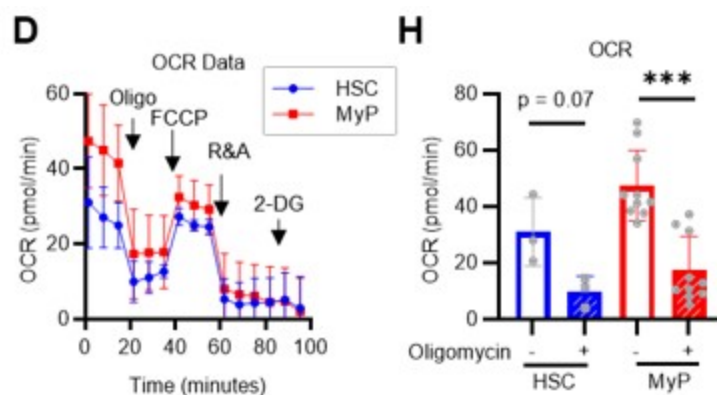


FIG.3 A-C. As mentioned previously, the flux analyses should be integrated with data using other energy sources. If cycling HSCs are less dependent to OXPHOS, what happen if you inhibit OXPHOS in 5-FU condition? Since the authors are linking OXPHOS inhibition and upregulation of Glycolysis to increase proliferation, do HSCs proliferate more when treated with oligomycin?

First, please see our response to comments 3 and 5 regarding the first part of this comment about the flux analysis of other energy sources. According to the analysis using the GO-Ateam2 system, stressed HSCs change the efficiency of glucose utilization (accelerated glycolytic ATP production) rather than other energy sources. The change in ATP concentration after OXPHOS inhibition for 5-FU-treated HSCs is shown in Figures 4C and E, suggesting that the activity of OXPHOS itself does not increase. HSCs after oligomycin treatment and HSCs after 5-FU treatment are similar in that they activate glycolytic ATP production. However, inhibition of OXPHOS did not induce the proliferation of HSCs (original Figure S1K). This suggests that proliferation activates glycolysis and not that activation of the glycolytic system induces proliferation. This similarity and dissimilarity of glycolytic activation upon proliferation and OXPHOS inhibition is discussed in the Discussion section (page 16–17, lines 505–515).

- FIG.4 shows that in vivo administration of radiolabeled glucose especially marks metabolites of TCA cycle and Glycolysis. The authors interpret enhanced anaerobic glycolysis, but I am not sure this is correct; if more glycolysis products go in the TCA cycle, it might mean that HSC start engaging mitochondrial metabolism. What do the authors think about that?*

Thank you for pointing this out. We believe that the data are due to two differences in the experimental features between in vivo (Figure S5) and in vitro (Figures 1 and S2) tracer analysis. The first difference is that in in vivo tracer analysis, unlike in vitro, all cells can metabolize U-13C6-glucose. Another difference is that after glucose labeling in vivo, it takes approximately 120–180 minutes to purify HSCs to extract metabolites, and processing on ice may result in a gradual progression of metabolic reactions within HSCs. As a result, in vivo tracer analysis may detect an increased influx of labeled carbon derived from U-13C6-glucose

into the TCA cycle over an extended period. However, it is difficult to interpret whether this influx of labeled carbon is derived from the direct influx of glycolysis or the re-uptake by HSCs of metabolites that have been metabolized to other metabolites in other cells. Meanwhile, as shown in Figure 4C using the GO-ATeam2 system, ATP production from mitochondria is not upregulated by 5-FU treatment. This suggests that even if the direct influx from glycolysis into the TCA cycle is increased, the rate of ATP production does not exceed that of glycolysis. Despite these technical caveats in interpretation, the results of in vivo and in vitro tracer analyses are considered essential. In particular, we consider the increased labeling of metabolites involved in glycolysis and nucleotide synthesis to be crucial. We have added a discussion of these points, including experimental limitations (page 17-18, lines 530-545).

1. FIG.4: the experimental design is not clear. Are BMNNCs stained and then put in culture? Is it 6-day culture or BMNNCs are purified at day 6 post 5FU? FIG-4B-C The difference between PBS vs 5FU conditions are the most significant; however, the effect of oligomycin in both conditions is the most dramatic one. From this readout, it seems that HSCs are more dependent on mitochondria for energy production both upon 5FU treatment and in PBS conditions.

We apologize for the incomplete description of the experimental details. The experiment involved dispensing freshly stained BMMNC with surface antigens into the medium and immediately subjecting them to flow cytometry analysis. For post-5-FU treatment HSCs, mice were administered with 5-FU (day 1), and freshly obtained BMMNCs were analyzed on day 6. The analysis of HSCs and progenitors was performed by gating each fraction within the BMMNC (original Figure S5A). We have added these details to ensure that readers can grasp these aspects more clearly (page S5, lines 155-158).

As pointed out by the reviewer, we understand that HSCs produce more ATP through OXPHOS. However, ATP production by glycolysis, although limited, is observed under steady-state conditions (post-PBS treatment HSC), and its reliance increases during the proliferation phase (post-5-FU treatment HSC) (original Figures 4B, D). Until now, discussions on energy production in HSCs have focused on either glycolysis or mitochondrial functions. However, with the GO-ATeam2 system, it has become possible for the first time to compare their contributions to ATP production and evaluate compensatory pathways. As a result, it became evident that while OXPHOS is the main source of ATP production, the reliance on glycolysis plastically increases in response to stress. This has led to a better understanding of HSC metabolism. These points are included in the Discussion as well (page 16, lines 479-488).

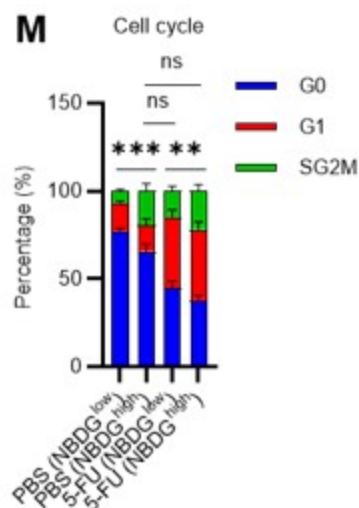
1. FIG.6H should be extended with cell cycle analyses. There are no differences between 5FU and ctrl groups. If 5FU induces HSCs cycling and increases glycolysis I would expect higher 2-NBDG uptake in the 5FU group. How do the authors explain this?

Thank you for your comments. In the original Figure 6H, we found that 2-NBDG uptake correlated with mPfkfb3 levels in both the 5-FU and PBS groups. mPfkfb3 levels remained low in the few HSCs with low 2-NBDG uptake in the 5-FU group.

In the revised manuscript, to directly relate glucose utilization to the cell cycle, we administered 2-NBDG to mice and fractionated HSCs at the 2-NBDG fluorescence level for cell cycle analysis. The results are shown below (revised Figure S1M). Notably, even in the PBS-treated group, HSCs with high 2-NBDG uptake were more proliferative than those with low 2-NBDG uptake and are comparable to HSCs after 5-FU treatment, although the overall population of HSCs that exited the G0 phase and entered the G1 phase increased after 5-FU treatment. The large differences in glucose utilization per cell cycle observed in both PBS/5-FU-treated groups suggest a direct link between HSC proliferation and glycolysis activation.

Descriptions of the above findings have been added to the Results and Discussion ((page 7, lines 208-214, page 20, lines 607-610).

Author response image 27.



1. In S7 the experimental design is not clear. What are quiescent vs proliferative conditions? What does it mean "cell number of HSC-derived colony"? Is it a CFU assay? Then you should show colony numbers. When HSCs proliferate, they need more energy thus inhibition of metabolism will impact proliferation. What happens if you inhibit mitochondrial metabolism with oligomycin?

Regarding the proliferative and quiescence-maintaining conditions, we have previously reported on these 8. In brief, these are culture conditions that maintain HSC activity at a high level while allowing for the proliferation or maintenance of HSCs in quiescence, achieved by culturing under fatty acid-rich, hypoxic conditions with either high or low cytokine concentrations. Analysis was performed after one week of culture, with the HSC number determined by flow cytometry based on the LSK-SLAM marker. While these are mentioned in the Methods section, we have added a description in the main text to highlight these points for the reader (page 7, lines 214-217).

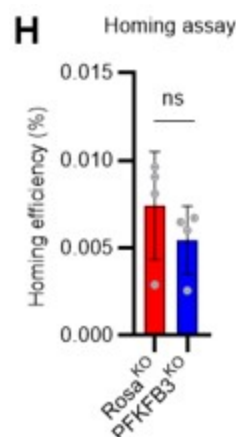
In vitro experiments with the oligomycin treatment of HSCs showed that OXPHOS inhibition activates the glycolytic system, but does not induce HSC proliferation (original Figure S1K). This suggests that proliferation activates glycolysis and not that activation of the glycolytic system induces proliferation. This similarity and dissimilarity of glycolytic activation upon proliferation and OXPHOS inhibition is discussed in the Discussion (page 16-17, lines 505-515).

1. In FIG 7 since homing of HSCs is influenced by the cell cycle state, should be important to show if in the genetic model for PFKFB3 in HSCs there's a difference in homing efficiency.

In response to the reviewer's comments, we knocked out PFKFB3 in HSPCs derived from Ubc-GFP mice, transplanted 200,000 HSPCs into recipients (C57BL/6 mice) post-8.5Gy irradiation, and harvested the bone marrow of recipients after 16 h to compare homing efficiency (revised Figure S10H). Even with the knockout of PFKFB3, no significant difference in homing efficiency was detected compared to the control group (Rosa knockout group). These results suggest that the short-term reduction in chimerism due to PFKFB3 knockout is not due to decreased homing efficiency or cell death by apoptosis (Figure 7K) but a transient delay in

cell cycle progression. We have added descriptions regarding these findings in the Results and Discussion sections (page 15, lines 470-471, page 19, lines 576-578).

Author response image 28.



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